

Distinct Modes of Presynaptic Inhibition of Cutaneous Afferents and Their Functions in Behavior

Highlights

- Two distinct mechanisms underlie PSI of cutaneous afferents and tactile sensitivity
- GABA_ARs underlie homotypic activity-dependent PSI of afferents
- Small-diameter afferents also engage heterotypic NMDAR-dependent PSI
- Presynaptic GABA_ARs, not NMDARs, are necessary for texture discrimination

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In Brief

Zimmerman et al. define distinct neurotransmitter and circuit mechanisms that gate the flow of tactile sensory information from the periphery into the spinal cord and characterize their roles in tactile sensitivity and behaviors.



Distinct Modes of Presynaptic Inhibition of Cutaneous Afferents and Their Functions in Behavior

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SUMMARY

Presynaptic inhibition (PSI) of primary sensory neurons is implicated in controlling gain and acuity in sensory systems. Here, we define circuit mechanisms and functions of PSI of cutaneous somatosensory neuron inputs to the spinal cord. We observed that PSI can be evoked by different sensory neuron populations and mediated through at least two distinct dorsal horn circuit mechanisms. Low-threshold cutaneous afferents evoke a GABA_A-receptor-dependent form of PSI that inhibits similar afferent subtypes, whereas small-diameter afferents predominantly evoke an NMDA-receptor-dependent form of PSI that inhibits large-diameter fibers. Behaviorally, loss of either GABA_A receptors (GABA_ARs) or NMDA receptors (NMDARs) in primary afferents leads to tactile hypersensitivity across skin types, and loss of GABA_ARs, but not NMDARs, leads to impaired texture discrimination. Post-weaning age loss of either GABA_ARs or NMDARs in somatosensory neurons causes systemic behavioral abnormalities, revealing critical roles of two distinct modes of PSI of somatosensory afferents in adolescence and throughout adulthood.

INTRODUCTION

Presynaptic inhibition (PSI) of synaptic inputs to early sensory processing areas of the CNS is believed to control gain and contrast and filter irrelevant sensory cues in a context- and state-dependent manner. In the visual system, PSI of bipolar cell terminals contributes to center surround inhibition of ganglion cell receptive fields for enhanced visual contrast (Buldyrev and Taylor, 2013). In the olfactory system, PSI of primary sensory neuron input to the olfactory bulb controls temporal contrast, thus enabling pattern separation and odor discrimination and learning (Gödde et al., 2016; Raccuglia et al., 2016). In the somatosensory system, PSI of cutaneous sensory neuron inputs to the spinal cord may underlie general gain control and acuity (Eccles et al., 1963b; Frank and Fuortes, 1957; Rudomin and

Schmidt, 1999), although its functions and mechanisms remain incompletely understood despite over 70 years of research.

The anatomical substrate of PSI of both cutaneous and proprioceptive afferents of the somatosensory system is believed to be inhibitory axoaxonic synapses that form upon sensory neuron terminals within the spinal cord and brainstem. In the spinal cord dorsal horn, the central projections of C- and A δ -fiber cutaneous sensory afferents, and some A β afferents, terminate within complex synaptic structures termed glomeruli, where GABAergic axoaxonic as well as presumptive dendroaxonic synaptic inputs form onto them (Todd, 1996). Thus, dorsal horn GABAergic interneurons, whose activity may be controlled by descending modulatory- or sensory-evoked signals, release GABA to activate ionotropic GABA receptor chloride channels (GABA_ARs) present on primary sensory neuron terminals. Due to the high intracellular chloride concentration in primary sensory neurons, maintained by NKCC1 throughout development and into adulthood (Alvarez-Leefmans et al., 1988; Kanaka et al., 2001), activation of GABA_ARs on sensory neuron terminals leads to efflux of Cl⁻ ions, which depolarizes the terminal. This process is termed primary afferent depolarization (PAD). Electrical stimulation of sensory afferents or supraspinal centers, or mechanical stimulation of the skin, can evoke PAD in a number of afferent subtypes (Andersen et al., 1962; Carpenter et al., 1963; Eccles et al., 1963b; Jänig et al., 1968; Jiménez et al., 1987), and GABA_AR antagonists diminish low-threshold cutaneous and proprioceptive afferent-evoked PAD in the cat, rat, turtle, and mouse (Eccles et al., 1963a; Russo et al., 2000; Shreckengost et al., 2010). PAD underlies PSI by paradoxically reducing transmitter release from afferent terminals (Eccles et al., 1963b; Lidieth, 2006), which may involve one or more hypothesized mechanisms: (1) shunting inhibition, caused by Cl⁻ efflux leading to diminished action potential (AP) height and thus transmitter release; (2) inactivation of voltage-gated Na⁺ channels, thereby diminishing AP height and transmitter release; and (3) inactivation of voltage-gated Ca²⁺ channels in terminals, reducing transmitter release (Rudomin and Schmidt, 1999). Experimentally, PAD can be measured in individual sensory neuron fibers or, more conveniently, with an extracellular recording electrode placed on the dorsal root and recording a dorsal root potential (DRP), which reflects back-propagating PAD in primary sensory neuron axons (Eccles et al., 1963b; Lidieth, 2006).

Although GABA_AR activation is widely considered a crucial step leading to PSI of primary somatosensory neurons, several

early studies reported incomplete block of sensory afferent- or supraspinal-evoked PAD by GABA_AR antagonists, with some *in vivo* recordings implying that inhibition of GABA_ARs can enhance evoked DRPs (Benoist et al., 1972; Besson et al., 1971). Moreover, *in vitro* findings have suggested a role for presynaptic ionotropic glutamate receptors in inhibition of afferent transmission in the dorsal horn (Bardoni et al., 2004; Lee et al., 2002; Russo et al., 2000), and embryonic loss of NMDA receptors (NMDARs) in primary afferents leads to heat and mechanical hypersensitivity in adult mice (Pagadala et al., 2013). Both small- and large-diameter dorsal root ganglion (DRG) neurons express GABA_AR (Paul et al., 2012) and ionotropic glutamate receptor (Lee et al., 2002; Marvizón et al., 2002) subunits, and PAD can be elicited in both small- and large-caliber fibers (Whitehorn and Burgess, 1973). Yet earlier studies were limited in defining the sensory neuron subtypes that evoke PAD, circuit mechanisms governing activity-dependent PAD, and functions of PSI of cutaneous afferents in tactile behaviors.

Here, we used mouse genetic tools, optogenetics, electrophysiology, pharmacology, and behavioral approaches to probe the mechanisms and functions of PSI of cutaneous sensory neuron inputs to the dorsal horn. We found that, in addition to GABA_ARs present on the central terminals of both nociceptors and low-threshold mechanoreceptors (LTMRs), NMDARs also function presynaptically on sensory neuron terminals in the dorsal horn to control sensory inputs. Although GABA_AR-mediated PSI functions in a predominantly homotypic manner, NMDAR-mediated PSI is predominantly heterotypic. Furthermore, distinct populations of dorsal horn interneurons can elicit these two forms of PSI. Disruption of either GABA_AR- or NMDAR-dependent forms of PSI in adolescent mice, through selective ablation of genes encoding obligatory subunits of neurotransmitter receptors in primary somatosensory neurons, augments tactile sensitivity across multiple skin types. In contrast, loss of GABA_ARs, but not NMDARs, results in deficits in texture recognition. Interestingly, reducing either form of PSI leads to alterations in cognitive behaviors. Thus, two modes of PSI, which are evoked by different sensory neuron populations and mediated through distinct dorsal horn circuit mechanisms, collaborate in the control of mechanosensation and behavior.

RESULTS

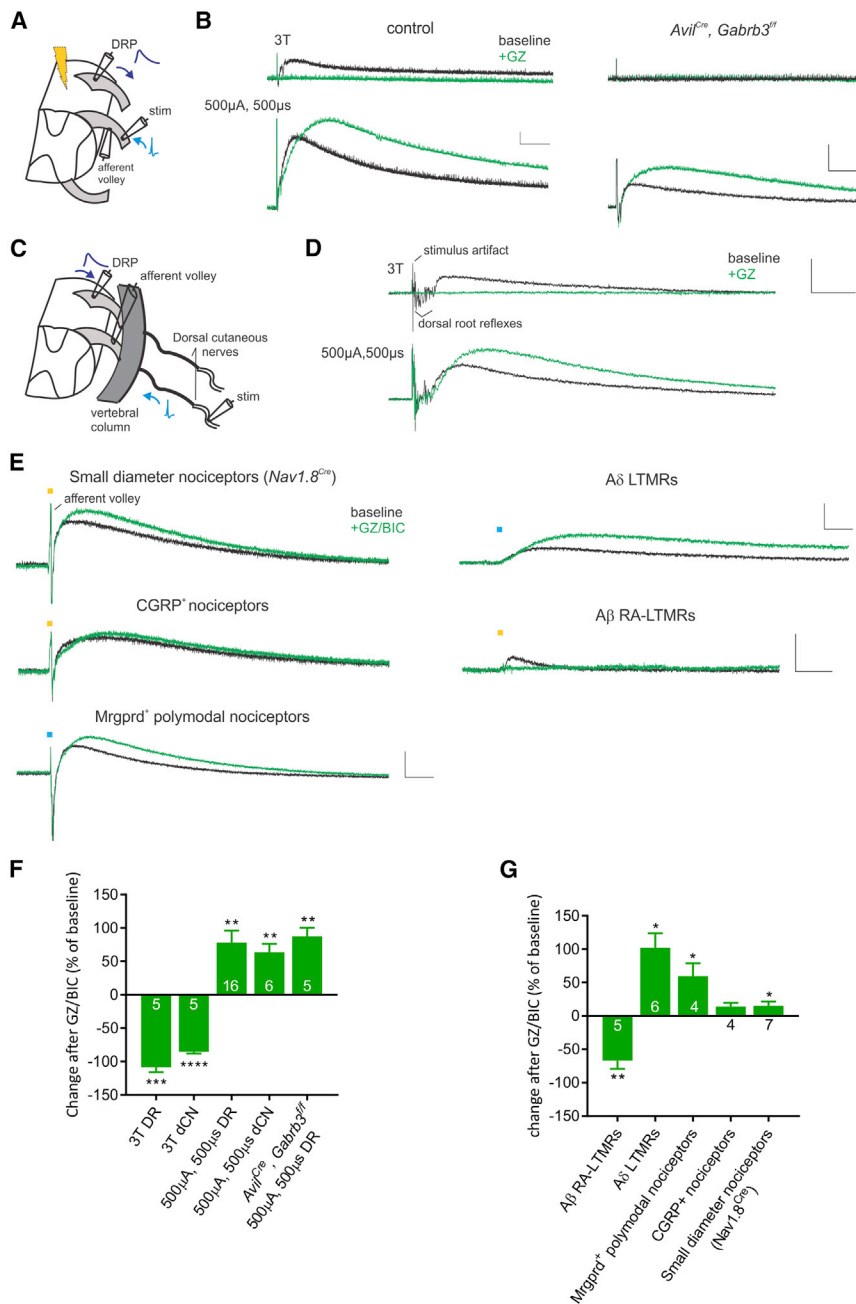
Two Mechanistically Distinct Forms of PSI of Sensory Input to the Dorsal Horn

As a measure of PSI of sensory input to the spinal cord, we determined the extent of PAD using an *ex vivo* hemisectioned spinal cord preparation. By positioning a suction electrode on one dorsal root, a back-propagating wave of depolarization in sensory neuron axons, or DRP, was recorded following electrical stimulation of the adjacent dorsal root (DR) (Figure 1A). This preparation allowed us to record afferent-evoked DRPs and restrict the site of action of applied drugs to the spinal cord. We also recorded from the DR proximal to the stimulation site to measure the minimum current necessary to evoke afferent volleys. Recordings were done in postnatal day 14–18 mice, an age by

which inhibitory circuits in the spinal cord have matured to an adult-like state (Baccei and Fitzgerald, 2004) and when stimulating DRs at or below 3× threshold (3T) (7.5–24 μA; 100 μs) exclusively recruits A-fibers (Figure S1; Torsney and MacDermott, 2006). Stimulating the lower thoracic DRs at 3T, we observed a low threshold-evoked DRP that was completely blocked by the GABA_AR inhibitors gabazine (5–10 μM) or bicuculline (10–20 μM; Figure 1B). Yet when the DR was stimulated at high stimulus intensity (500 μA; 500 μs), which recruits both A- and C-fibers, the evoked DRP was not blocked by GABA_AR inhibitors but was instead enhanced, likely due to disinhibited spinal circuits. To directly assess the role of GABA_ARs within sensory neurons for low-threshold, afferent-evoked DRPs, we genetically eliminated GABA_ARs exclusively in somatosensory neurons by conditionally deleting GABRB3, an obligate GABA_AR subunit for inhibitory transmission in these neurons (Nguyen and Nicoll, 2018; Orefice et al., 2016), using *Avil^{Cre}; Gabrb3^{fl/fl}* mice. As predicted, the low-threshold-evoked DRP was completely absent in *Avil^{Cre}; Gabrb3^{fl/fl}* mice; however, high-intensity-evoked DRPs were present and, in fact, further enhanced by GABA_AR antagonists (Figures 1B and 1F). No sex-specific differences were observed in any DRP measurements (Figure S1A).

In order to assess cutaneous sensory-neuron-evoked responses, the preparation was adapted to stimulate only the dorsal cutaneous nerve (Figure 1C), which exclusively innervates hairy skin. When the dorsal cutaneous nerve was stimulated at 3T, a gabazine-sensitive DRP was elicited (Figures 1D and 1F), similar to that observed following DR stimulation. Also similar to DR stimulation, when the dorsal cutaneous nerve was stimulated at high intensity, the evoked DRP was enhanced by gabazine. These results, together with previously published work (e.g., De Koninck and Henry, 1994; Shreckengost et al., 2010), suggest that GABA_ARs are integral to low-threshold cutaneous afferent evoked PAD and are consistent with EM findings indicating that large-caliber fibers receive axoaxonic GABAergic synaptic inputs (Todd, 1996). Moreover, these findings indicate that high-intensity electrical stimulation of either the DR or dorsal cutaneous nerve recruits an additional, GABA_AR-independent form of PAD.

We next asked which sensory neuron subtypes evoke GABA_AR-dependent and GABA_AR-independent modes of PSI. Because electrical stimulation of both the DR and the dorsal cutaneous nerve cumulatively activates A α , A β , A δ , and C-fiber sensory neuron fibers at increasing stimulus intensities, this mode of stimulation is insufficient to define the sensory neuron subtypes involved. Therefore, we used an optogenetic approach to selectively activate defined sensory neuron subsets while measuring light pulse-evoked DRPs. Using *Nav1.8^{Cre}; Avil^{FlpO}* (for small diameter DRG neurons), *CGRP^{FlpO}; Avil^{Cre}* (CGRP⁺ peptidergic nociceptors), *Mrgd^{Cre}* (*Mrgpr*⁺ polymodal nociceptors), *TH^{2A-CreER}; Avil^{FlpO}* (C-LTMRs), *TrkB^{CreER}* (A δ -LTMRs), or *Ret^{CreER}; Avil^{FlpO}* (A β RA-LTMRs) mice crossed with an appropriate Cre and/or FlpO recombinase-dependent channelrhodopsin (*Rosa26^{LSL-ChR2}*) or ReaChR (*Rosa26^{LSL-FSF-ReaChR}* or *Rosa26^{FSF-ReaChR}*) mouse line, we systematically tested whether optogenetic activation of select sensory neuron subtypes is sufficient to evoke a DRP and, if so, whether the DRP is dependent



on GABA_ARs (Figures 1E and 1G). Strikingly, activation of all small diameter (Nav1.8⁺) DRG neurons, CGRP⁺ peptidergic nociceptors, Mrgprd⁺ polymodal nociceptors, or Aδ-LTMRs evoked a DRP that was largely independent of GABA_ARs. We were unable to evoke a DRP following optogenetic stimulation of C-LTMRs, even though optogenetically evoked afferent volleys could be observed (data not shown). Although there may be a recording bias of afferent types with extracellular suction electrodes leading to negative findings, C-LTMRs are numerous in the DRG (Li et al., 2011), suggesting that activity in this subtype of LTMR engages little PAD under these experimental conditions. In addition, whether any of these small- and medium-

Aδ-LTMRs evoke a predominantly GABA_AR-independent form of PSI.

Presynaptic NMDARs Are Present on LTMR Terminals and Mediate High-Threshold PAD

Because C-fiber sensory neuron subtypes and Aδ-LTMRs can evoke PSI through a non-canonical, GABA_AR-independent mechanism, we next undertook a series of pharmacological experiments to define the receptor system involved. Nicotinic cholinergic receptor (nAChR) antagonists, glycine receptor antagonists, kainate receptor antagonists, metabotropic glutamate antagonists, and GABA_B receptor antagonists all failed to

Figure 1. There Are at Least Two Mechanistically Distinct Forms of Sensory-Neuron-Evoked PAD

(A) Recording configuration for dorsal root (DR) or optogenetic stimulation.

(B) Effect of GABA_AR antagonist gabazine (GZ) (10 μM) on low-threshold (≤3T) and high-intensity (500 μA; 500 μs) DR evoked DRPs in control and *Avil^{Cre}; Gabrb3^{fl/fl}* mice. Note the loss of low-threshold-evoked DRPs in *Avil^{Cre}; Gabrb3^{fl/fl}* mice.

(C) Recording configuration for dorsal cutaneous nerve stimulation.

(D) Similar to DR stimulation, GZ blocks 3T dorsal cutaneous nerve evoked DRP but enhances the high-intensity-electrical-stimulation-evoked DRP.

(E) Example traces of optogenetically evoked DRPs at baseline and in the presence of GABA_AR antagonists GZ (5–10 μM) or bicuculline (BIC) (10–20 μM). Blue and yellow bars denote light pulse timing. The following crosses were used: small-diameter nociceptors (*Nav1.8^{Cre}*; *Avil^{FlpO}*; *Rosa26^{LSL-FSF-ReaChR}*); CGRP nociceptors (*Avil^{Cre}*; *CGRP^{FlpO}*; *Rosa26^{LSL-FSF-ReaChR}*); Mrgprd⁺ polymodal nociceptors (*Mrgprd^{Cre}*; *Rosa26^{LSL-ChR2}*); Aδ-LTMRs (*TrkB^{CreER}*; *Rosa26^{LSL-ChR2}*); and Aβ RA-LTMRs (*Ret^{CreER}*; *Avil^{FlpO}*; *Rosa26^{LSL-FSF-ReaChR}*).

(F and G). Summary of the effects of GABA_AR antagonists on electrically (F) and optogenetically evoked DRPs (G).

All bar graphs in figures are represented as mean ± SEM. Asterisks above bar graphs denote statistical significance (****p < 0.0001; ***p < 0.001; **p < 0.01; *p < 0.05). Scale bars represent 0.05 mV, 100 ms.

diameter subtypes are also able to evoke a GABA_AR-dependent form of PAD is unclear, as the dominant GABA_AR-independent form of PAD that is disinhibited by GABA_AR antagonists would likely occlude this response. In contrast, optogenetic activation of Aβ RA-LTMRs, using *Ret^{CreER}*; *Avil^{FlpO}*; *Rosa26^{LSL-FSF-ReaChR}* mice, elicited a DRP that was predominantly GABA_AR-dependent (Figures 1E and 1G). Thus, Aβ-LTMRs evoke GABA_AR-dependent PSI, whereas C-fiber sensory neuron subtypes and

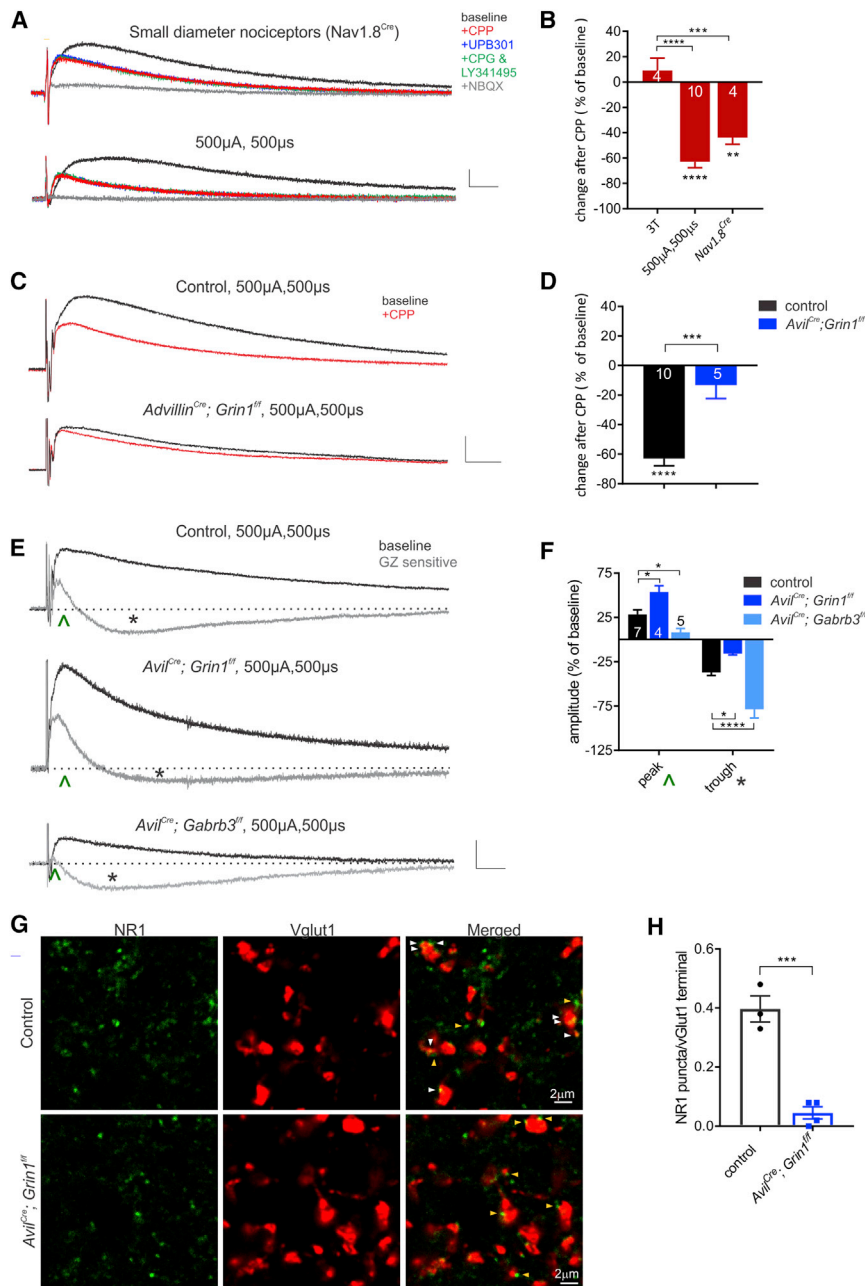


Figure 2. Presynaptic NMDARs Contribute to High-Threshold PAD

(A) Optogenetic activation of small-diameter afferents (Nav1.8^{Cre}; Avil^{Flpo}; Rosa26^{LSL-FSF-ReaChR}) and high-intensity electrical stimulation of the dorsal-root-evoked DRPs in baseline conditions and with the sequential addition of an NMDAR antagonist (CPP; 20 µM), kainate receptor antagonist (UPB301; 10 µM), metabotropic glutamate receptor antagonists (MCPG, 50 µM and LY341495, 50 µM), and AMPA and kainate receptor antagonist (NBQX; 10 µM). Only the combination of ionotropic glutamate receptor inhibitors blocks both types of DRP.

(B) Summary of the effects of CPP (20 µM) on low-threshold electrical, high-intensity electrical, and small-diameter optogenetically evoked DRPs. Only the latter two types of evoked DRPs are inhibited by CPP (one-way ANOVA; $F(2,13) = 24.75$; $p < 0.0001$). Asterisks denote Tukey's post hoc test.

(C) Example of the effect of CPP on DRPs in control mice and Avil^{Cre}, Grin1^{ff} mutants.

(D) Summary of CPP-evoked changes on 500 µA, 500 µs evoked DRPs. Asterisks below bar graph denote statistical significance from baseline; asterisk above denotes two-tailed Student's *t* test; $p = 0.0003$.

(E) Control and GZ subtracted traces of 500 µA, 500 µs dorsal root stimulation in control, Avil^{Cre}; Grin1^{ff}, and Avil^{Cre}; Gabrb3^{ff} mutants. ^ denotes gabazine-sensitive peak; asterisk denotes gabazine-enhanced trough.

(F) Normalized GZ-sensitive peak and trough amplitude. Two-way ANOVA, genotype $F(2,26) = 30.61$, $p < 0.0001$; component $F(1,26) = 206.6$, $p < 0.0001$. Dunnett's multiple comparison post hoc test compares each mutant to control mice, denoted by asterisks.

(G) Immunohistochemistry of lamina III-IV for NR1 and myelinated afferent terminals marked by VGlut1. White arrows denote presynaptic NR1 puncta, and yellow arrows denote postsynaptic NR1 puncta in control mice and Avil^{Cre}; Grin1^{ff} mutants.

(H) Average number of NR1 puncta per VGlut1 afferent terminal. Unpaired *t* test, $p = 0.0005$.

**** $p < 0.0001$; *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

diminish high-threshold-evoked DRPs (Figures 2A and S1). In fact, both nAChR antagonists and glycine receptor antagonists modestly enhanced high-threshold-evoked DRPs (Figure S2), although the site of action underlying this facilitation is presently unclear. Conversely, the ionotropic NMDAR and AMPA receptor antagonists CPP (3-(2-Carboxypiperazin-4-yl)-propyl-1-phosphonic acid) and NBQX (2,3-Dioxo-6-nitro-1,2,3,4-tetrahydrobenzo[f]quinoxaline-7-sulfonamide), respectively, diminished high-threshold DRPs evoked either optogenetically, using Nav1.8^{Cre} to express either ChR2 or ReaChR, or electrically using high-intensity DR stimulation (Figure 2A). Because both primary afferents and interneurons of spinal cord polysynaptic PSI cir-

cuits employ glutamatergic transmission, it was not possible to determine whether the glutamate receptor blockers acted directly on sensory neuron terminals to diminish PAD. Therefore, we turned to a gene ablation approach to determine whether NMDARs present on the central terminals of primary sensory neurons themselves are responsible for C-fiber- and Aδ-LTMR-evoked DRPs. For this, we used Avil^{Cre}; Grin1^{ff} mice to conditionally delete the obligatory NMDAR subunit NR1 exclusively in primary somatosensory neurons and performed DRP measurements. High-threshold-evoked DRPs were diminished, and importantly, the CPP sensitivity of evoked responses was occluded in Avil^{Cre}; Grin1^{ff} mice (Figures 2C and S2). These

findings indicate the NMDARs expressed in primary sensory neurons contribute to high-threshold, sensory-neuron-evoked PAD.

The pharmacological and genetic deletion analyses suggested that GABA_AR-dependent and NMDAR-dependent forms of PAD summate to form the high-threshold-evoked DRP. In order to more clearly measure these two separate components, we defined the extent of GABA_AR-sensitive PSI by subtracting DRP traces generated in the presence and absence of gabazine using *Avil^{Cre}*; *Gabrb3^{flf}* and *Avil^{Cre}*; *Grin1^{flf}* mice, which lack GABA_ARs or NMDARs in primary sensory neurons, respectively. High-intensity electrical stimulation evoked an early-onset, GABA_AR-dependent DRP and a later onset, longer lasting DRP that was not blocked but rather disinhibited by GABA_AR antagonists. The early GABA_AR-dependent component was absent in *Avil^{Cre}*; *Gabrb3^{flf}* mutants although, in contrast, the longer latency GABA_AR-independent component was diminished in *Avil^{Cre}*; *Grin1^{flf}* mutants (Figures 2E and 2F). The identity of the receptor on sensory neuron terminals mediating the non-GABAergic PAD remaining in *Avil^{Cre}*; *Grin1^{flf}* mutants remains unknown, but our pharmacology experiments and related *in vitro* work (Lee et al., 2002) implicate presynaptic AMPA receptors.

Moreover, consistent with the notion that presynaptic NMDARs (preNMDARs) mediate GABA_AR-independent PSI, immunohistochemical analysis of the NR1 subunit of NMDARs and VGlut1, expressed in large-diameter sensory afferent terminals (Abraira et al., 2017), revealed that NMDARs are localized to ~40% of VGlut1⁺ terminals in the dorsal horn. Similar co-localization was observed between NR1 and sensory neuron terminal synapses, using *Avil^{Cre}*; *Rosa26^{LSL-syaptophysin-tdTomato}* reporter mice in which presynaptic boutons of DRG sensory neurons are labeled with tdTomato (Figure S3). These immunohistochemical measures indeed reflect NMDARs on sensory neuron terminals because co-localization of NR1 and VGlut1 was absent in *Avil^{Cre}*; *Grin1^{flf}* mutants, in which NR1 is eliminated in sensory neurons, but not in spinal cord neurons (Figures 2G and 2H). In addition, *in situ* hybridization experiments confirmed the presence of NMDAR transcripts in both large- and small-diameter DRG sensory neurons (Figure S3). These findings are consistent with prior work showing that preNMDARs are localized to primary afferent terminals in the rat spinal cord by electron microscopy (EM) (Liu et al., 1994), and NMDA application to the isolated spinal cord induces a Mg-sensitive DRP and reduces DR-evoked AMPAR currents in the dorsal horn (Bardoni et al., 2004). Taken together, our findings indicate that NMDARs are localized to primary sensory neuron terminals, where they contribute to C- and Aδ-fiber-evoked PSI. Thus, there exist at least two mechanistically distinct forms of PSI of sensory input to the dorsal horn: a GABA_AR-mediated form evoked by Aβ-LTMRs and a NMDAR-mediated form that is evoked by peptidergic and non-peptidergic C-fiber sensory neuron subtypes and Aδ-LTMRs.

GABA_AR-Dependent PAD and NMDAR-Dependent PAD Are Activated by Different Excitatory Interneurons of the Dorsal Horn

Because distinct sensory neuron subtypes promote either GABA_AR- or NMDAR-dependent PSI, we next tested the idea that different populations of dorsal horn interneurons are

involved. In our previous work (Abraira et al., 2017), we identified *CCK^{Cre}*-labeled interneurons as a large excitatory population comprising ~30% of interneurons within the LTMR recipient zone of the dorsal horn. Interestingly, selective activation of *CCK^{Cre}*-labeled interneurons using optogenetic stimulation of *CCK^{Cre}*; *Lbx1^{Flpo}*; *Rosa26^{LSL-FSF-ReaChR}* or *CCK^{Cre}*; *Rosa26^{LSL-ChR2}* mice (Figure 3A) evoked a strong DRP, and this was largely abolished by GABA_AR antagonists when recorded in young (<postnatal day 8 [P8]) mice (Figures 3B and 3C). When *CCK^{Cre}*-labeled interneuron-evoked DRPs were examined in P14–P18 mice (the same age recorded for experiments in Figures 1 and 2), GABA_AR antagonists selectively diminished the early DRP (Figures 3B and 3C), suggesting that *CCK^{Cre}*-labeled excitatory interneurons include those that engage GABA_AR-dependent PSI, likely through a GABAergic interneuron intermediate. We also assessed the involvement of transient VGlut3⁺ excitatory interneurons of the dorsal horn, which are implicated in mechanical allodynia (Peirs et al., 2015). VGlut3⁺ interneuron terminals span the more ventral nociceptive and entire LTMR recipient zone, and these excitatory interneurons receive inputs from both small- and large-diameter afferents (Figure 3H; Peirs et al., 2015). Although a portion of transient VGlut3⁺ interneurons overlap with *CCK^{Cre}*-labeled interneurons (Figure S4), selective activation of transient VGlut3⁺ interneurons, by optogenetic stimulation of the DR entry zone of *VGlut3^{Cre}*; *Lbx1^{Flpo}*; *Rosa26^{LSL-FSF-ReaChR}* mice, evoked a strong DRP that was entirely independent of GABA_ARs but partially dependent on NMDARs (Figures 3D and 3E). Thus, as with Aβ-LTMRs, C-fibers, and Aδ-LTMRs, a subset of *CCK^{Cre}*-labeled interneurons and transient VGlut3⁺ interneurons engage distinct modes of PAD to mediate PSI in the dorsal horn.

Because AMPA-R antagonists had little effect on the transient VGlut3⁺ interneuron-evoked DRP (Figure 3D), we next asked whether this mode of interneuron-evoked DRP is due to direct actions onto afferent terminals. Although tetrodotoxin (TTX) (1 μM) dramatically reduced VGlut3 interneuron-evoked DRPs, and it eliminated the high-intensity DR stimulation-evoked DRP in these mice, as expected for a polysynaptic circuit, the addition of the K⁺ channel blocker 4-aminopyridine (4-AP) rescued only the light-evoked DRP (Figures 3F and 3G). Distinct from most optogenetic-assisted circuit mapping analyses, light-evoked DRPs partially remained in the presence of TTX alone, suggesting some light-evoked transmitter release, and at a high dose of 4-AP (1 mM), light-evoked DRPs often exceeded baseline values (data not shown). These findings suggest either monosynaptic input from transient VGlut3⁺ interneurons onto sensory neuron terminals or, because the vast majority of axoaxonic and a majority of presumptive dendroaxonic synapses in dorsal horn glomeruli are GABAergic (Todd, 1996), perhaps more likely an extrasynaptic mode of action. Consistent with both of these possibilities, labeled synaptic terminals of transient VGlut3⁺ interneurons using *VGlut3^{Cre}*; *Lbx1^{Flpo}*; *Rosa26^{LSL-FSF-syaptophysin-GFP}* mice (Niederkofler et al., 2016) were observed throughout the LTMR-recipient zone, often in close proximity to VGlut1⁺ terminals of sensory neurons (Figure 3H). Whether synaptic or extrasynaptic, activation of the transient VGlut3⁺ interneurons evokes PAD in a manner that does not appear to depend on an additional, intermediate dorsal

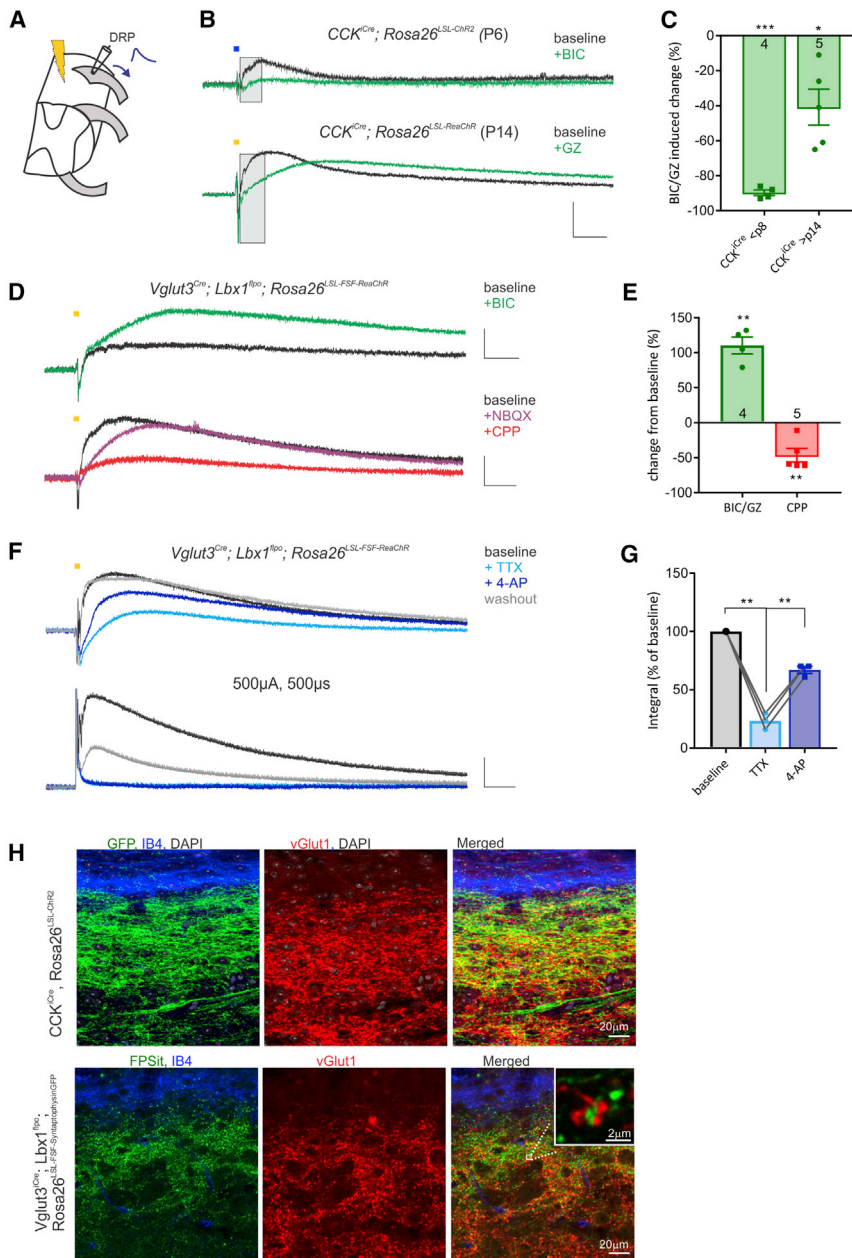


Figure 3. GABA_AR PAD and NMDA-R PAD Are Engaged by Different Populations of Dorsal Horn Excitatory Interneurons

(A) Recording configuration denoting light stimulation of the dorsal root entry zone.

(B) CCK^{Cre}-labeled interneuron-induced DRPs in neonatal or juvenile mice are completely or partially blocked by GABA_AR antagonists, respectively. Shaded region demarcates area under the DRP calculated in (C).

(C) Composite data, integrating the area under the filtered DRP up to its peak in both neonatal and juvenile mice. Asterisks denote statistical significance from no effect.

(D) Transient VGlut3⁺ interneuron (labeled using VGlut3^{Cre}, Lbx1^{Flo} mice)-activated DRPs depend on ionotropic glutamate receptors, particularly NMDARs, and are enhanced by GABA_AR antagonists.

(E) Composite data, integrating area under the full-filtered DRP for each antagonist.

(F) Application of tetrodotoxin (TTX) (1 μ M) greatly diminishes the transient VGlut3⁺ interneuron optogenetically evoked DRP, which is partially recovered by the addition of 600 μ M 4-AP. Electrically evoked DRPs were completely blocked by TTX and could not be recovered with the addition of 4-AP (400 μ M–1 mM).

(G) Composite data of the effect of TTX (1 μ M) and 4-AP (400–600 μ M) on optogenetically evoked DRPs. One-way ANOVA; $F(1.241, 2.482) = 307.2$; $p = 0.0011$. Asterisks denote Tukey's multiple comparison post hoc test. Physiology scale bars denote 0.05 mV, 100 ms.

(H) Immunohistochemistry of transient VGlut3⁺ interneurons and CCK^{Cre}-labeled interneurons. (Upper) Sagittal sections from CCK^{Cre}, Rosa26^{SL-ChR2} mouse counterstained for IB4 and VGlut1 are shown; (lower) sagittal sections from VGlut3^{Cre}, Lbx1^{Flo}, Rosa26^{SL-FSF-Synaptophysin-GFP} mouse counterstained for VGlut1 and IB4 are shown. Inset is high magnification of close proximity of transient VGlut3 interneuron terminals to VGlut1 presumptive afferent terminals.

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

horn interneuron. Taken together, these findings support the existence of a glutamatergic form of PSI mediated by the transient VGlut3⁺ interneurons of the dorsal horn.

GABA_ARs and NMDARs in Primary Somatosensory Neurons Differentially Control Cutaneous Tactile Sensitivity and Acuity

Our observations reveal the existence of at least two distinct receptor mechanisms and dorsal horn circuit motifs controlling PSI of sensory neuron terminals. Therefore, we next sought to define the roles of these two modes of PSI in behavioral responses to natural cutaneous stimuli. To avoid compensation and non-cell-autonomous effects from developmental disruption of

NMDAR- and GABA_AR-mediated transmission, we conditionally ablated either *Gabrb3* or *Grin1* in somatosensory neurons of post-weaning mice using the conditional *Gabrb3* and *Grin1* alleles and the somatosensory neuron selective, tamoxifen-sensitive allele *Avil^{CreER}* (Lau et al., 2011). Thus, *Avil^{CreER}; Gabrb3^{fl/fl}* mice and *Avil^{CreER}; Grin1^{fl/fl}* mice and their Cre-negative control littermates were treated with tamoxifen (1 mg/day) for 5 consecutive days, beginning at P28. This treatment regimen achieved ~80% reduction of *Gabrb3* mRNA and ~40% reduction of *Grin1* mRNA in DRGs of experimental mice (Figure 4A), reflecting a reduction, but not elimination, of these obligate receptor subunits and likely results in an underestimation of the requirement of these receptors. Male and female control and experimental

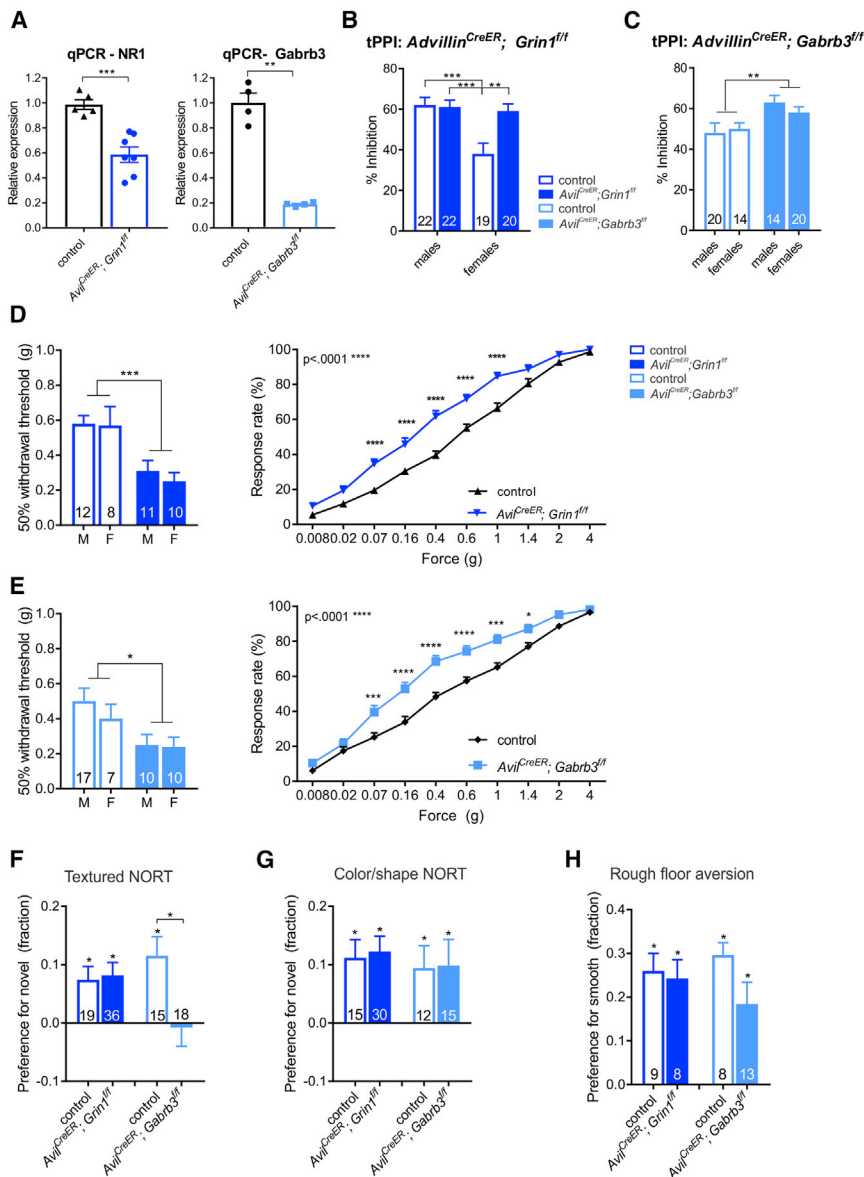


Figure 4. GABA_ARs and NMDA-Rs Are Both Required in Adult Somatosensory Afferents for Normal Tactile Sensitivity

(A) qPCR of mRNA knockdown in both *Avil^{CreER}* conditional knockout models. Unpaired two-tailed Student's *t* test with Welch's correction; *p* = 0.0003 for *Grin1* and *p* = 0.0018 for *Gabrb3*.

(B) tPPI, 50 ms ISI, *Avil^{CreER}; Grin1^{fl/fl}*. Two-way ANOVA: interaction *F*(1,79) = 7.437, *p* = 0.0079; sex *F*(1,79) = 10.39, *p* = 0.0018; genotype *F*(1,79) = 6.146, *p* = 0.0153. Tukey's multiple comparisons test denoted by asterisks.

(C) tPPI, 50 ms ISI, *Avil^{CreER}; Gabrb3^{fl/fl}*. Two-way ANOVA, asterisks denote main effect of genotype (*F*(1,64) = 8.618; *p* = 0.0046); no main effect of sex or interaction.

(D and E) Von Frey thresholds and response rates for *Avil^{CreER}; Grin1^{fl/fl}* (D) and *Avil^{CreER}; Gabrb3^{fl/fl}* (E).

Two-way ANOVA of threshold, main effect of genotype denoted by asterisks. *Avil^{CreER}; Grin1^{fl/fl}*: *F*(1,37) = 16.05, *p* = 0.0003; *Avil^{CreER}; Gabrb3^{fl/fl}*: *F*(1,40) = 6.937, *p* = 0.0119. For response rates, a two-way repeated-measures ANOVA was performed, and the main effect of genotype is noted by the *p* value in the upper left and asterisks denoting Sidak's multiple comparisons post hoc test. No main effect of sex or interaction for either genotype is shown.

(F–H) Textured NORT (F), shape NORT (G), and rough floor aversion tests (H) for both *Grin1^{fl/fl}* and *Gabrb3^{fl/fl}* crosses. Asterisks above bar graphs denote preference significantly different than zero. Unpaired two-tailed Student's *t* test per cross; *p* = 0.0122 for *Avil^{CreER}; Gabrb3^{fl/fl}* mice textured NORT.

*****p* < 0.0001; ****p* < 0.001; ***p* < 0.01; **p* < 0.05.

mice were subjected to a battery of behavioral tests at 8–11 weeks of age, with experimenters blind to genotype. As a measure of hairy skin tactile sensitivity, we used a tactile prepulse inhibition assay (tPPI), where an acoustic startle stimulus is preceded by a light air puff delivered to the back hairy skin of animals with varying inter-stimulus intervals (Orefice et al., 2016). For this assay, the more sensitive a mouse is to the air puff, the more the air puff prepulse attenuates the acoustic startle response (i.e., increased PPI). We observed that female, but not male, *Avil^{CreER}; Grin1^{fl/fl}* mice and, as predicted from previous work (Orefice et al., 2016), both female and male *Avil^{CreER}; Gabrb3^{fl/fl}* mice exhibit enhanced tPPI compared to controls, indicative of hairy skin hypersensitivity (Figures 4B and 4C).

Both *Avil^{CreER}; Grin1^{fl/fl}* and *Avil^{CreER}; Gabrb3^{fl/fl}* mice also exhibited glabrous skin hypersensitivity, as assessed using von Frey filament stimulation of hindpaw glabrous skin. Thresholds

and response rates to filament stimulation were abnormal in the conditional knockouts, and no difference between males and females was observed (Figures 4D and 4E). When estrus checking female mice for other behavioral measures not included in this study, we noticed an additional potential tactile sensitivity phenotype; many mice exhibited vigorous responses following contact with the genital skin area. We therefore generated a genital skin sensitivity scale (see Videos S1 and S2 and STAR Methods). *Avil^{CreER}; Grin1^{fl/fl}* females, but not *Avil^{CreER}; Gabrb3^{fl/fl}* females, showed markedly increased sensitivity to contact with the genital skin area compared to controls (Figure S6A). *Avil^{CreER}; Grin1^{fl/fl}* mice, but not *Avil^{CreER}; Gabrb3^{fl/fl}* mice, also exhibited a mild hypersensitivity to heat pain, as assessed using the Hargreaves test (Figures S6B and S6C). In contrast to a prior study (McRoberts et al., 2011), hyposensitivity to inflammatory pain was not observed in either *Avil^{CreER}; Grin1^{fl/fl}* or *Nav1.8^{Cre}; Grin1^{fl/fl}* male mice (Figures S6D and S6E). Other pain and temperature assays were not assessed, reflecting an interesting future direction.

PSI is implicated in lateral inhibition, which in other sensory systems contributes to spatial and temporal acuity (Buldyrev

and Taylor, 2013; Gödde et al., 2016; Raccuglia et al., 2016). Therefore, we also assessed texture discrimination in controls and mutant mice using a textured novel object recognition task (textured NORT) (Orefice et al., 2016). In this behavioral paradigm, control mice typically spend more time exploring a novel block compared to a familiar block. Two versions of NORT were done: one in which blocks differ in texture but are identical in shape and color (textured NORT) and a control NORT in which blocks differ in shape and color but have the same texture (color and shape NORT). As we previously observed with heterozygous *Avil^{CreER}; Gabrb3^{fl/+}* mice (Orefice et al., 2016), homozygous *Avil^{CreER}; Gabrb3^{fl/fl}* mutant mice did not exhibit preference for novel textured objects, although they did show normal novelty preference for objects that differ in shape and color (Figures 4F and 4G). These findings confirm that GABA_ARs in somatosensory neurons are crucial for texture discrimination. In contrast, *Avil^{CreER}; Grin1^{fl/fl}* mice exhibited a similar preference for textured objects to controls (Figure 4F). Thus, NMDARs in sensory neurons may be dispensable for this behavior. Finally, both *Avil^{CreER}; Grin1^{fl/fl}* mice and *Avil^{CreER}; Gabrb3^{fl/fl}* mice exhibited aversion to rough textured floors, comparable to their littermate controls (Figure 4H), indicating that avoidance of an aversive texture is not dependent on either mode of PSI.

Although it is likely that more dramatic behavioral phenotypes would be observed with a more complete gene deletion in somatosensory neurons, these findings indicate that loss of either NMDARs or GABA_ARs in primary afferents causes hypersensitivity across trunk hairy skin and hindpaw glabrous skin, and loss of NMDARs, but not GABA_ARs, in sensory neurons causes genital skin hypersensitivity and modest temperature hypersensitivity. Moreover, consistent with a role for PSI of A β -LTMRs in tactile acuity, a reduction of GABA_ARs, but not NMDARs, in sensory neurons leads to deficits in fine texture discrimination.

Presynaptic GABA_ARs and NMDARs Both Function in Large-Diameter Sensory Neurons for Sensory Neuron-Evoked PSI

The finding that two mechanistically distinct forms of PSI differentially contribute to cutaneous tactile sensitivity and acuity raises the question of which classes of sensory neurons are modulated by the two forms of PSI. To address this, we used the *ex vivo* DRP recording preparation and mice harboring the *Nav1.8^{Cre}* allele and the *Gabrb3* and *Grin1* conditional alleles to abolish either GABA_AR- or NMDAR-mediated PSI exclusively in small-to-medium-diameter sensory neurons, which are labeled with *Nav1.8^{Cre}*. We found that high-intensity electrical-stimulation-evoked DRPs were diminished in magnitude in *Nav1.8^{Cre}; Gabrb3^{fl/fl}* mice, whereas low-threshold-evoked DRPs were unaffected in these mutants (Figures 5A and 5B). Moreover, both low-threshold- and high-intensity-evoked DRPs exhibited similar sensitivity to gabazine in *Nav1.8^{Cre}; Gabrb3^{fl/fl}* and control mice (Figures 5C and 5D). These findings indicate that activation of large-diameter sensory neuron subtypes (with 3T DR stimulation), including A β -LTMRs, promotes GABA_AR-mediated PSI predominantly of large-diameter sensory neuron inputs to the dorsal horn (i.e., those not labeled with *Nav1.8^{Cre}*). In contrast, high-intensity electrical stimulation of afferents promotes at least two forms of PSI, including GABA_AR-

mediated PSI of small-to-medium-diameter afferents. To test whether the latter reflects small-diameter afferent-evoked inhibition of other small-diameter afferents, we selectively activated small-diameter fibers while simultaneously silencing GABA_ARs in the same afferents (with *Nav1.8^{Cre}; Gabrb3^{fl/fl}; Rosa26^{LSL-ReaChR}* mice). We found that the light-activated DRP was diminished in *Nav1.8^{Cre}; Gabrb3^{fl/fl}; Rosa26^{LSL-ReaChR}* mice when compared to light-activated control (*Nav1.8^{Cre}; Rosa26^{LSL-ReaChR}*) mice, with similar gabazine sensitivity (Figure S5). Thus, although the dominant form of PAD evoked by small-diameter afferents is GABA_AR independent (Figure 1), a coinciding weaker form of PAD is evoked by small-diameter afferents and induces GABA_AR-dependent PAD of other small-diameter afferents.

In related experiments, NR1 was deleted in small-diameter afferents using *Nav1.8^{Cre}; Grin1^{fl/fl}* mice, and the pharmacological sensitivity of evoked DRPs in these mice was compared to those observed in control mice and *Avil^{Cre}; Grin1^{fl/fl}* mice, which lack NMDARs in all primary sensory neurons. Interestingly, CPP application attenuated the high-threshold-evoked DRPs in *Nav1.8^{Cre}; Grin1^{fl/fl}* mice to an extent comparable to that observed in control mice; this attenuation of CPP sensitivity was not observed in pan-DRG *Grin1* knockout mice (Figures 5E and 5F). This indicates a minimal contribution of NMDARs on small-diameter afferents to sensory-neuron-evoked PAD. Together with the results shown in Figure 1, these findings indicate that NMDAR-dependent PAD is evoked exclusively by small-to-medium-caliber afferents and expressed mainly in large-diameter afferents, which are not labeled with *Nav1.8^{Cre}*. These findings support a model in which presynaptic GABA_ARs (preGABA_ARs) predominantly mediate homotypic (A β →A β , C/A δ →C/A δ) PSI, whereas preNMDARs primarily mediate heterotypic (C/A δ →A β) PSI of cutaneous afferents (Figure 5G).

NMDARs on DCN-Projecting A β -LTMRs Regulate Hairy Skin Sensitivity, and GABA_ARs on Both Small-to-Medium-Diameter Neurons and DCN-Projecting A β -LTMRs Affect Glabrous Skin Sensitivity

Our physiological findings indicate that both GABA_ARs and NMDARs function within presynaptic terminals of large-diameter neurons, and GABA_ARs also function in C/A δ afferents to mediate sensory-evoked PSI. Therefore, we asked whether deletion of GABA_ARs or NMDARs in either large- or small and medium-diameter sensory neuron subtypes recapitulates any of the tactile hypersensitivity phenotypes observed in the pan-DRG *Gabrb3* and *Grin1* mutant mice. To delete *Gabrb3* or *Grin1* exclusively in small- and medium-diameter afferents, we again used *Nav1.8^{Cre}; Gabrb3^{fl/fl}* and *Nav1.8^{Cre}; Grin1^{fl/fl}* mice. To delete *Gabrb3* or *Grin1* in large-diameter neurons, including A β -LTMRs, but not in small- and medium-diameter DRG neurons, we took advantage of the fact that only A β /A α afferents project via the dorsal column to the dorsal column nuclei (DCN) of the brainstem (Bai et al., 2015). Thus, we injected a FlpO-dependent Cre adeno-associated virus (AAV2/9-fDIO-Cre-GFP; Penzo et al., 2015) into the rostral dorsal column of *Avil^{FlpO}* mice to direct Cre-dependent recombination exclusively in DCN-projecting afferents (Figure 6A). This labeling strategy

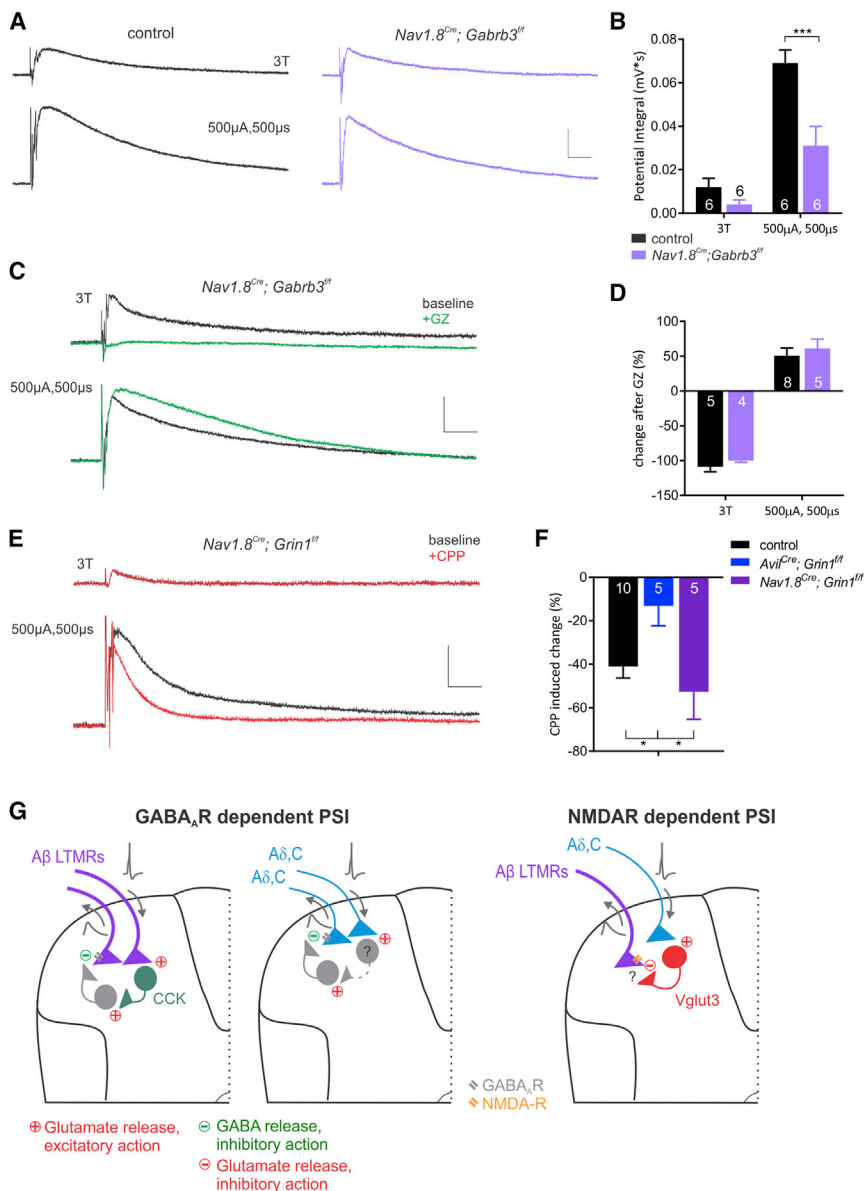


Figure 5. GABA_ARs Mediate PSI of Both Large-Diameter and Small-Diameter Sensory Neurons, whereas NMDARs Mediate PSI Predominantly in Large-Diameter Neurons

(A) Sample traces of same-day-recorded littermate control and *Nav1.8^{Cre}; Gabrb3^{fl/fl}* mutant DRPs.

(B) Composite data of DRP integrals, two-way ANOVA: interaction of stimulus intensity and genotype; $F(1,20) = 6.461$; $p = 0.0194$. Asterisks denote Sidak's multiple comparisons test, compared within each stimulus intensity.

(C) Example traces from *Nav1.8^{Cre}; Gabrb3^{fl/fl}* mutant DRPs evoked by stimulating an adjacent dorsal root at stimulus intensity noted and with the addition of 10 μ M gabazine (GZ).

(D) Composite data of GZ sensitivity. Control and *Nav1.8^{Cre}; Gabrb3^{fl/fl}* mutant mice have bidirectional GZ modulation of evoked DRPs (two-way ANOVA; no main effect of genotype or interaction with stimulus intensity).

(E) Example traces from *Nav1.8^{Cre}; Grin1^{fl/fl}* mutant DRPs evoked by dorsal root stimulation using the stimulation intensities noted and with the addition of CPP. Note marked CPP-induced reduction of the high-intensity DRP remains.

(F) Composite data comparing control, *Nav1.8^{Cre}; Grin1^{fl/fl}*, and *Avil^{Cre}; Grin1^{fl/fl}* mice (repeated from Figure 2). One-way ANOVA: $F(2,12) = 5.228$; $p = 0.0233$. Asterisks below bars denote Dunnett's multiple comparison post hoc test. All scale bars denote 0.05 mV, 100 ms. *** $p < 0.001$; * $p < 0.05$.

(G) Diagrams of three afferent activity-dependent modes of PSI. Left most spinal cord dorsal horn represents GABA_AR dependent PSI of large-diameter cutaneous afferents. Activity in A β -LTMRs can lead to activation of a subset of CCK^{Cre}-labeled excitatory interneurons, which in turn activate GABAergic interneurons and induce GABA_AR-dependent PAD in predominantly other A β subtypes. Middle spinal cord represents GABA_AR-dependent PSI of small-diameter afferents, induced by activity in small-diameter afferents. Currently unidentified interneuron(s) are colored gray. The right spinal cord dorsal horn represents NMDAR-dependent PSI, whereby activity in C and A δ afferents predominantly inhibits A β -LTMRs through transient VGlut3⁺ excitatory interneurons. The anatomical substrate of this form of PSI is unknown but does not require additional synapses.

was tested using *Avil^{FlpO}; Rosa26^{LSL-tdTomato}* (Ai14) mice. Indeed, tdTomato expression was observed in ~26% of NF200⁺ caudal thoracic DRG neurons, with GFP labeling ~50% of these neurons, whereas neither tdTomato nor GFP labeling was observed in neurons lacking NF200 or in Ai14 control mice (Figure 6B). Moreover, tdTomato⁺ terminals were observed exclusively within the VGlut1⁺ LTMR recipient zone of the spinal cord, indicating transduction of large-diameter, but not small-diameter, sensory neuron subtypes by this approach. In addition, no somatic labeling of the spinal cord was observed in these mice. Next, the efficiency of NR1 deletion in fDIO-Cre-GFP virus-injected *Avil^{FlpO}; Grin1^{fl/fl}* mice was assessed using RNAscope. In both GFP⁺/Cre⁺ and Cre⁺-only neurons, NR1 expression was undetectable (Figure 6C). This approach thus ablates genes in

a large subset of DCN-projecting sensory neurons, which includes A β -LTMR subtypes, but not in A δ -LTMRs, C-fiber sensory neuron subtypes, or spinal cord neurons.

We next injected AAV2/9-fDIO-Cre-GFP into the rostral dorsal column of *Avil^{FlpO}; Grin1^{fl/fl}* and *Avil^{FlpO}; Gabrb3^{fl/fl}* mice and their control littermates at p28 and used these mice along with *Nav1.8^{Cre}; Grin1^{fl/fl}* and *Nav1.8^{Cre}; Gabrb3^{fl/fl}* mice and their control littermates for tactile sensitivity behavioral measurements at 8–10 weeks of age. In a subset of mice, the efficiency of viral transduction was assessed by measuring the extent of GFP and NF200 co-localization using immunohistochemistry (IHC) following completion of behavioral analyses. This IHC analysis confirmed efficient and selective transduction in *Avil^{FlpO}* mice, but not control mice (Figure 6D). Using this paradigm, we

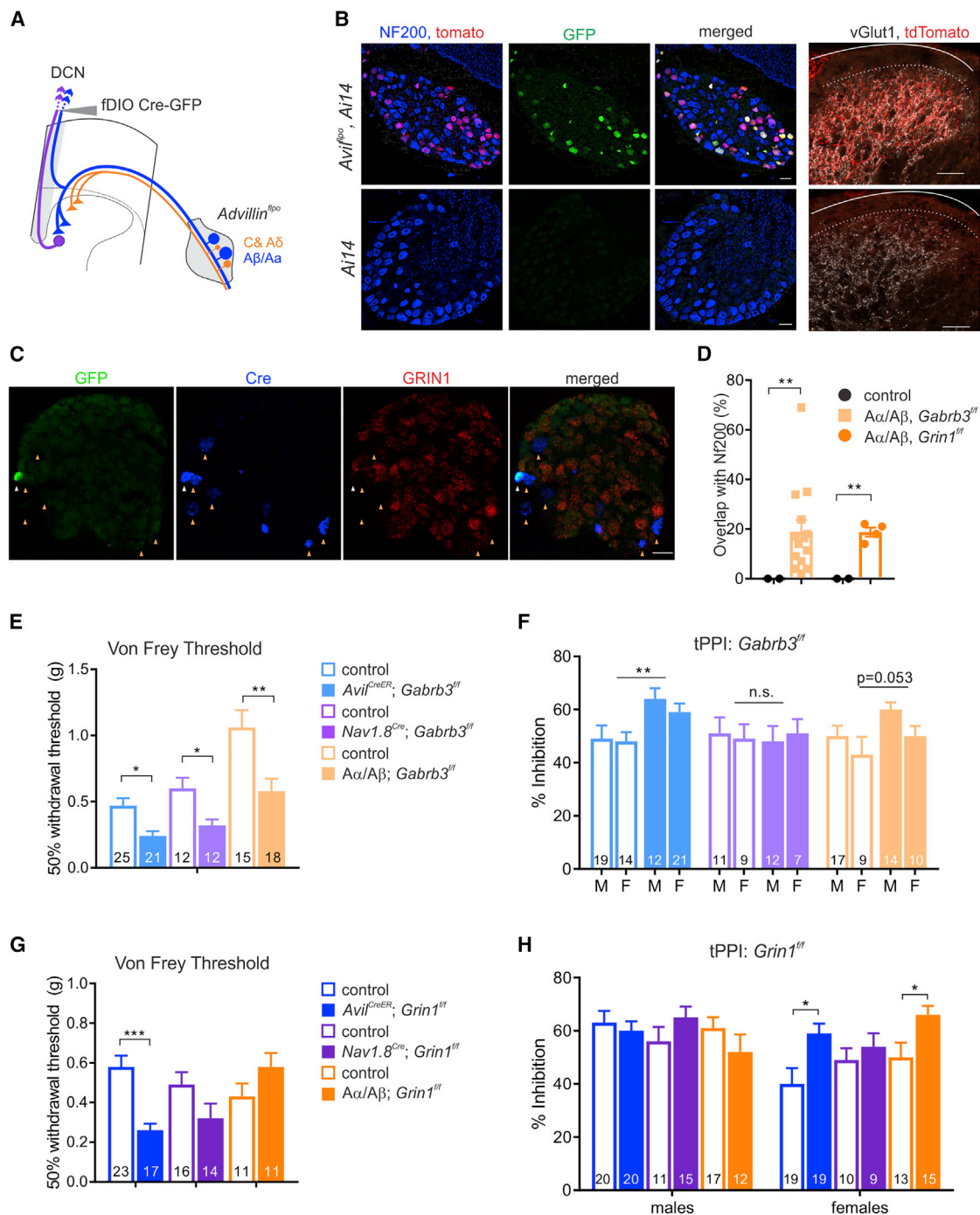


Figure 6. NMDAR-Mediated PAD of Aα/Aβ Afferents Underlies Hair Skin Hypersensitivity, and GABA_AR-Mediated PAD of Both C/Aδ Afferents and Aα/Aβ Afferents Underlies Glabrous Skin Hypersensitivity

(A) Diagram of viral strategy to conditionally delete receptors only in DCN-projecting A-fibers.

(B) Characterization of viral strategy of AAV2/9-fDIO-Cre-GFP in *Avil^{FlpO}; Rosa26^{LSL-tdTomato}* (Ai14) mice and control (Ai14 only) mice in DRG and spinal cord dorsal horn (right most). GFP expression and Cre-driven tdTomato expression was observed only in NF200⁺ neurons in the DRG of *Avil^{FlpO}* mice. Scale bars denote 20 μm for DRG and 50 μm for spinal cord.

(C) RNAscope for Cre and Grin1 expression in *Avil^{FlpO}; Grin1^{fl}* mice and immunohistochemistry for GFP expression. White arrowhead denotes Cre⁺/GFP⁺ soma; yellow arrowheads note Cre⁺/GFP⁻ cells.

(D) Labeling efficiency of virus-injected mice previously run on behavioral assays.

(legend continued on next page)

observed hypersensitivity to von Frey filaments applied to the hindpaw in both *Nav1.8^{Cre}; Gabrb3^{fl/fl}* mutants and the *Avil^{FlpO}; Gabrb3^{fl/fl}* mutants injected with AAV2/9-fDIO-Cre-GFP in the DCN (Figure 6E). These findings of glabrous skin hypersensitivity are consistent with the physiological findings implicating GABA_ARs in PSI of both A β afferents and C/A δ neurons. Conversely, when GABA_ARs were deleted exclusively in A β /A α afferents, hairy skin hypersensitivity was not observed, although there was a trend in that direction (Figure 6F); this negative finding may reflect incomplete deletion using the dorsal column nuclei viral ablation paradigm (Figure 6D). Consistent with this possibility, tPPI was unperturbed in *Nav1.8^{Cre}; Gabrb3^{fl/fl}* mice. Thus, GABA_ARs in both A β -fiber and C/A δ -fiber subtypes modulate glabrous skin sensitivity, and GABA_ARs in large-diameter afferents, afferents unlabeled by either strategy, or a combination of subtypes modulate hairy skin sensitivity.

In contrast, neither *Nav1.8^{Cre}; Grin1^{fl/fl}* mice nor *Avil^{FlpO}; Grin1^{fl/fl}* mice injected with AAV2/9-fDIO-Cre-GFP into the rostral dorsal column recapitulated the glabrous skin hypersensitivity to von Frey stimulation observed in mice lacking *Grin1* in all DRG neurons (Figure 6G). However, deletion of NMDARs in DCN-projecting A β /A α afferents led to a similar female-specific hairy skin tactile hypersensitivity phenotype as observed in *Avil^{CreER}; Grin1^{fl/fl}* mice (Figure 6H). This suggests that preNMDARs function in DCN-projecting A β -LTMRs to control hairy skin sensitivity in female mice.

Loss of GABA_AR and NMDAR-Mediated PSI in Young Adult Mice Results in Changes in Anxiety-like and Exploratory Behaviors

Because post-weaning age loss of NMDARs and GABA_ARs in somatosensory neurons leads to tactile hypersensitivity and our previous findings demonstrated that impaired tactile information processing and reactivity during development contribute to anxiety and social interaction abnormalities in adult mice (Orefice et al., 2016), the consequences of adolescent (P28) disruption of the receptor systems underlying PSI on complex behaviors were next tested. 2- or 3-month-old *Avil^{CreER}; Grin1^{fl/fl}* and *Avil^{CreER}; Gabrb3^{fl/fl}* mice and their littermate controls (which received tamoxifen treatments starting at P28) were subjected to open field and elevated plus maze tests and their startle amplitude and habituation to startle assessed from acoustic PPI trial data. Overall, both sets of mutants had similar behavioral abnormalities, distinct from their control littermates. Both *Avil^{CreER}; Grin1^{fl/fl}* and *Avil^{CreER}; Gabrb3^{fl/fl}* mutant mice exhibited hyper-locomotion during the open field test, covering greater distances (Figures 7A and 7B). Similarly, *Avil^{CreER}; Grin1^{fl/fl}* and *Avil^{CreER}; Gabrb3^{fl/fl}* mice ex-

hibited increased exploratory behavior, spending more time in the center of the open field chamber (Figures 7A and 7C) and displaying less aversion to the open arms of the elevated plus maze (Figure 7D), with female mutants showing a more severe phenotype (Figure S7). Both *Avil^{CreER}; Grin1^{fl/fl}* and *Avil^{CreER}; Gabrb3^{fl/fl}* mice also exhibited an increased acoustic startle reflex and decreased short-term habituation to acoustic startle (Figures 7E and 7F). Moreover, 2- or 3-month-old *Grin1* mutants exhibited aggressive behavior, including fratricide (data not shown). Unexpectedly, both *Avil^{CreER}; Grin1^{fl/fl}* and *Avil^{CreER}; Gabrb3^{fl/fl}* mice exhibited alterations in acoustic PPI (Figure S7), with deficits in *Avil^{CreER}; Grin1^{fl/fl}* mice being more pronounced. As spiral ganglion afferents are not labeled by *Avil^{CreER}* P28 tamoxifen treatment (data not shown), this phenotype may suggest cross-sensory modality compensation for aberrant tactile filtering. Together, these findings indicate that alterations of the receptor systems underlying both modes of PSI in adolescent mice leads to general, overt changes in exploratory and anxiety-like behaviors in the adult.

DISCUSSION

Here, we report that the spinal cord dorsal horn contains at least two distinguishable circuit motifs governing presynaptic inhibition of somatosensory afferents. These two PSI circuits are driven by different sensory neuron inputs, they are comprised of distinct dorsal horn excitatory interneuron populations, and they engage distinct receptor mechanisms on select afferent fiber subtype endings in the dorsal horn. The two modes of PSI play similar roles in some tactile behaviors yet differential roles in others, thus providing insight into the scope of behavioral influence of peripherally and centrally driven PSI of cutaneous afferents. Moreover, changes in exploratory and anxiety-like behaviors resulting from loss of somatosensory neuron GABA_ARs and NMDARs implicate both modes of PSI of somatosensory afferents as essential for general mouse behavior and late-stage cognitive development.

Different Somatosensory Neuron Subtypes and Dorsal Horn Circuits Underlie Two Distinct Modes of PSI

Our findings reveal the presence of both homotypic, GABA_AR-dependent and predominantly heterotypic, NMDAR-dependent modes of PSI of afferent input to the dorsal horn. The homotypic nature of GABA_AR-dependent PSI is consistent with prior work suggesting that, for proprioceptors and glabrous skin-innervating A β -LTMRs (Jänig et al., 1968), activity in one afferent type recruits GABAergic inhibition preferentially of similar or the same afferent types. In addition, our finding that activation

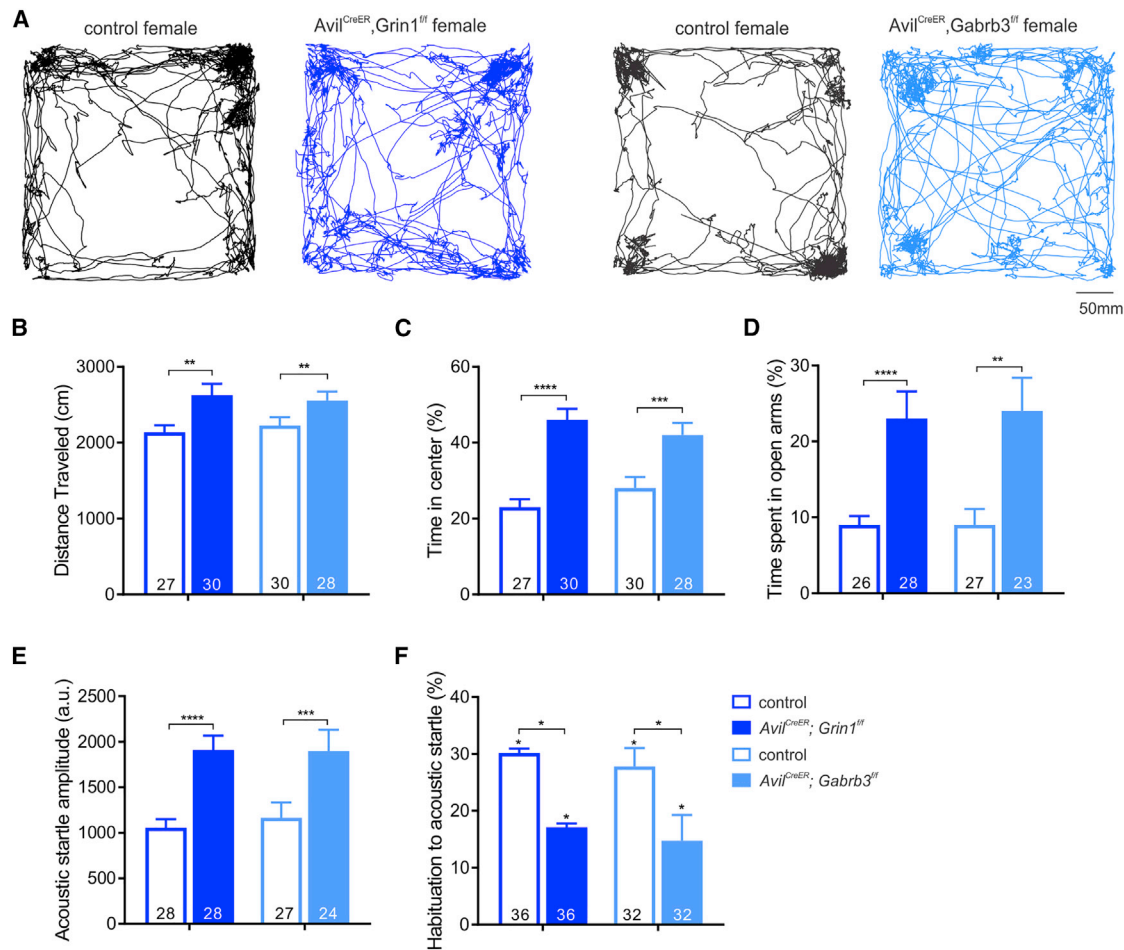
(E) 50% withdrawal threshold for Von Frey test for *Gabrb3^{fl/fl}* crosses. Two-way ANOVA was run separately on each cross; asterisks denote main effect of genotype. *Nav1.8^{Cre}; Gabrb3^{fl/fl}* main effect of genotype: $F(1,26) = 6.136$, $p = 0.0201$; *Avil^{FlpO}; Gabrb3^{fl/fl}* main effect of genotype: $F(1,36) = 8.506$, $p = 0.0061$.

(F) tPPI with 50 ms ISI for all three *Gabrb3^{fl/fl}* crosses; M labels male data and F labels female data. No significant effects of sex or interactions between sex and genotype for any cross are shown.

(G) Von Frey thresholds for *Grin1^{fl/fl}* crosses; asterisks denote main effect of genotype.

(H) tPPI with 50 ms ISI for all three *Grin1^{fl/fl}* crosses. Two-way ANOVA interaction between sex and genotype for *Avil^{FlpO}; Grin1^{fl/fl}* mutants; $F(1,61) = 6.244$; $p = 0.0132$. For each cross, Sidak's multiple comparison post hoc test was performed within each sex and denoted with asterisks. *Avil^{CreER}; Gabrb3^{fl/fl}* and *Avil^{CreER}; Grin1^{fl/fl}* data are replotted from data in Figure 4.

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.



of a subset of *CCK^{icre}*-labeled dorsal horn excitatory interneurons can recruit this GABA_AR-dependent form of PAD is consistent with a tri-synaptic circuit motif activated or modulated by both afferent inputs and descending projections from cortex and brainstem. Moreover, our behavioral and physiological findings support the idea that a GABA_AR-dependent form of PSI also engages small-diameter afferents in a homotypic manner, yet the interneuron(s) involved in this form of PAD are currently unidentified. Additional studies assessing the necessity and sufficiency of the dorsal horn interneurons that mediate each form of PAD are warranted.

In contrast to GABA_AR-dependent PSI, NMDAR-dependent PSI is uniquely activated by small- and medium-caliber afferents, including A δ -LTMRs, and acts predominantly on large-caliber afferents, including A β -LTMRs. The transient VGlut3⁺

population of dorsal horn interneurons, which play a role in mechanical allodynia after nerve injury or inflammation (Peirs et al., 2015), appears to directly engage somatosensory afferents to activate NMDAR-dependent PSI. It is interesting that GABA_AR antagonists enhance or “disinhibit” NMDAR-dependent PSI, especially in light of prior work implicating decreased dorsal horn GABAergic tone and GABA_AR-dependent PSI in mechanical allodynia in rodent inflammatory and neuropathic pain models (Malan et al., 2002; Zhang et al., 2008). These findings together may suggest the existence of cross-talk between spinal GABAergic inhibition and the NMDAR-dependent form of PSI, leading us to speculate that an altered balance of inhibitory mechanisms may contribute to the genesis and/or maintenance of mechanical allodynia associated with chronic pain states.

Functions of PSI in Mechanosensation and Behavior

P28 deletion of *Grin1* and *Gabrb3* (present study) mimics hypersensitivity characterized with embryonic deletion of these subunits (Orefice et al., 2016; Pagadala et al., 2013), indicating critical roles of these receptors in regulating mechanosensation in adult mice. Moreover, although our findings implicate GABA_AR- and NMDAR-mediated PSI in controlling tactile sensitivity in overlapping but distinct skin regions, only GABA_AR-mediated PSI is implicated in texture discrimination. These findings are consistent with a general, modulatory role of both GABA_AR- and NMDAR-mediated PSI in controlling gain in the tactile sensory system and a unique, obligatory role for GABA_AR-dependent PSI in texture discrimination. An appealing explanation for this difference is that GABA_AR-dependent PSI underlies lateral inhibition of glabrous skin-innervating A β -LTMR terminals, thus providing a substrate for controlling acuity.

Our findings also implicate preNMDARs in DCN-projecting A β /A α afferents for controlling hairy skin sensitivity, and preGABA_ARs in both DCN-projecting A β /A α afferents and small-caliber afferents for controlling glabrous skin sensitivity. Because the DCN viral knockout strategy used to ablate receptor genes is incomplete and variable (Figure 6H) and probably spares receptors expressed in at least some hairy-skin A β SAI-LTMRs of lumbar DRGs (Bai et al., 2015), our approach almost assuredly underestimates the roles of preGABA_ARs and preNMDARs in A β -LTMRs in the behavioral measures reported. Likewise, although both forms of PSI are clearly diminished in DRG conditional knockouts, the incomplete reductions may reflect preferential loss in some afferent subtypes and likely leads to an underestimation of the roles of these presynaptic receptors. It is also possible that NMDARs and/or GABA_ARs have a peripheral locus of action (e.g., Iwashita et al., 2012) that may contribute to the behavioral phenotypes observed or that centrally evoked PSI contributes to other sensory functions not assessed using our battery of behavioral tests, such as thermal preference, cold, and itch sensation.

An additional consideration pertains to the origin of centrally evoked PSI, either supraspinally or spinally driven, and whether it engages GABA_AR-mediated PSI, NMDAR-mediated PSI, or both. Interestingly, afferent-evoked DRPs remain fully sensitive to NMDAR antagonists when small-caliber afferents are devoid of *Grin1* (Figure 5), indicating that NMDARs contribute minimally, if at all, to sensory-neuron-evoked PAD of small-to-medium-diameter sensory neuron subtypes. Yet most, if not all, DRG sensory neurons express *Grin1* mRNA (Figure S2; Y. Zheng and D.D.G., unpublished data), and electrophysiological and EM evidence supports the conclusion that NMDARs are broadly expressed in DRG neurons (Li et al., 2004; Liu et al., 1994). Thus, although our findings indicate that sensory neuron-evoked NMDAR-dependent PSI plays a minor role in controlling glutamate release from nociceptors under normal conditions, it is possible that preNMDARs on C/A δ fiber afferents contribute to centrally evoked PSI. PreNMDARs may also function in nociceptors under pathological conditions because they are upregulated in rat models of neuropathic pain and opiate tolerance, where they may function by

enhancing glutamate release from nociceptors instead of inhibiting it (Yan et al., 2013; Zeng et al., 2006).

A notable outcome of the present study is that manipulations that diminish PSI of somatosensory afferents in adolescent mice, beginning at P28 and causing tactile hypersensitivity, lead to changes in exploratory and anxiety-like behaviors in 8- to 11-week-old mice. These phenotypes contrast with those seen when *Gabrb3* was deleted embryonically, and they also differ from heterozygous conditional *Gabrb3* deletion (*Avil^{CreER}; Gabrb3^{f/+}* mutants) beginning at P28 (Orefice et al., 2016). Indeed, when *Gabrb3* was deleted in all sensory neurons during embryonic development, marked decreases in exploration and no change in startle amplitude or aPPI were observed, and P28 heterozygous deletion of *Gabrb3* did not lead to changes in these behavioral measures. These findings, taken together, provide additional support for the existence of a developmental switch in the dependency of PSI and tactile sensitivity for overt mouse behaviors (Orefice et al., 2016). It is also intriguing that the combined phenotypes of hyper-locomotion, decreased aPPI, and increased acoustic startle reflex, shown here for post-weaning age homozygous deletions of both *Gabrb3* and *Grin1*, have also been noted in mouse models of systemic NR1 hypofunction (Duncan et al., 2004), human mutations implicated in autism and schizophrenia (Didriksen et al., 2017), and adolescent social isolation models in rodents (Fone and Porkess, 2008). We thus propose that two mechanistically and functionally distinct modes of PSI of sensory inputs to the dorsal horn collaborate to control gain and discriminatory capabilities of the somatosensory system, deficiencies of which compromise an animal's perception of, and interaction with, the outside world.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- KEY RESOURCES TABLE
- CONTACT FOR REAGENT AND RESOURCE SHARING
- EXPERIMENTAL MODEL AND SUBJECT DETAILS
 - Animals
- METHOD DETAILS
 - Electrophysiology
 - Tissue Fixation and Immunohistochemistry
 - RT-qPCR
 - RNAscope
 - Image analysis
 - Dorsal Column Injections
 - Behavioral Assays
 - Elevated Plus Maze (EPM)
 - Open Field Test
 - Novel Object Recognition Test (NORT)
 - Prepulse Inhibition (PPI) Startle Assay
 - Von Frey Assay
 - Textured Floor Aversion Assay
 - Estrus Testing (Vaginal Cytology) and Scoring
 - Hargreaves Test
 - Formalin Sensitivity Test
 - Statistical Analysis

SUPPLEMENTAL INFORMATION

Supplemental Information includes seven figures and two videos and can be found with this article online at <https://doi.org/10.1016/j.neuron.2019.02.002>.

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AUTHOR CONTRIBUTIONS

A.L.Z. and D.D.G. conceived the study; A.L.Z. performed all electrophysiology experiments; A.L.Z., E.M.K., and R.Y.P. performed genetic manipulations and histological and behavioral experiments; A.T. generated and characterized the GABRB3 antibody; Q.Z. contributed to the analysis of VGLUT3 interneurons and qPCR; A.L.Z. developed software and analyzed data; and A.L.Z. and D.D.G. wrote the paper with editing from E.M.K. and input from all authors.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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REFERENCES

- Abraira, V.E., Kuehn, E.D., Chirila, A.M., Springel, M.W., Toliver, A.A., Zimmerman, A.L., Orefice, L.L., Boyle, K.A., Bai, L., Song, B.J., et al. (2017). The cellular and synaptic architecture of the mechanosensory dorsal horn. *Cell* 168, 295–310.e19.
- Alvarez-Leefmans, F.J., Gamiño, S.M., Giraldez, F., and Noguerón, I. (1988). Intracellular chloride regulation in amphibian dorsal root ganglion neurones studied with ion-selective microelectrodes. *J. Physiol.* 406, 225–246.
- Andersen, P., Eccles, J.C., and Sears, T.A. (1962). Presynaptic inhibitory action of cerebral cortex on the spinal cord. *Nature* 194, 740–741.
- Baccei, M.L., and Fitzgerald, M. (2004). Development of GABAergic and glycinergic transmission in the neonatal rat dorsal horn. *J. Neurosci.* 24, 4749–4757.
- Bai, L., Lehnert, B.P., Liu, J., Neubarth, N.L., Dickendesher, T.L., Nwe, P.H., Cassidy, C., Woodbury, C.J., and Ginty, D.D. (2015). Genetic identification of an expansive mechanoreceptor sensitive to skin stroking. *Cell* 163, 1783–1795.
- Bardoni, R., Torsney, C., Tong, C.-K., Prandini, M., and MacDermott, A.B. (2004). Presynaptic NMDA receptors modulate glutamate release from primary sensory neurons in rat spinal cord dorsal horn. *J. Neurosci.* 24, 2774–2781.
- Benoist, J.M., Besson, J.M., Conseiller, C., and Le Bars, D. (1972). Action of bicuculline on presynaptic inhibition of various origins in the cat's spinal cord. *Brain Res.* 43, 672–676.
- Besson, J.M., Rivot, J.P., Abdelmoumene, M., and Aleonard, P. (1971). Effects of gamma-hydroxybutyrate on segmental and cortical control of transmission of the afferent volley at spinal level. *Neuropharmacology* 10, 145–151.
- Buldyrev, I., and Taylor, W.R. (2013). Inhibitory mechanisms that generate centre and surround properties in ON and OFF brisk-sustained ganglion cells in the rabbit retina. *J. Physiol.* 591, 303–325.
- Carpenter, D., Lundberg, A., and Norrsell, U. (1963). Primary afferent depolarization evoked from the sensorimotor cortex. *Acta Physiol. Scand.* 59, 126–142.
- Chaplan, S.R., Bach, F.W., Pogrel, J.W., Chung, J.M., and Yaksh, T.L. (1994). Quantitative assessment of tactile allodynia in the rat paw. *J. Neurosci. Methods* 53, 55–63.
- Chen, J.T., Guo, D., Campanelli, D., Frattini, F., Mayer, F., Zhou, L., Kuner, R., Heppenstall, P.A., Knipper, M., and Hu, J. (2014). Presynaptic GABAergic inhibition regulated by BDNF contributes to neuropathic pain induction. *Nat. Commun.* 5, 5331. Published online October 30, 2014. <https://doi.org/10.1038/ncomms6331>.
- De Koninck, Y., and Henry, J.L. (1994). Prolonged GABAA-mediated inhibition following single hair afferent input to single spinal dorsal horn neurones in cats. *J. Physiol.* 476, 89–100.
- Didriksen, M., Fejgin, K., Nilsson, S.R.O., Birknow, M.R., Grayton, H.M., Larsen, P.H., Lauridsen, J.B., Nielsen, V., Celada, P., Santana, N., et al. (2017). Persistent gating deficit and increased sensitivity to NMDA receptor antagonism after puberty in a new mouse model of the human 22q11.2 micro-deletion syndrome: a study in male mice. *J. Psychiatry Neurosci.* 42, 48–58.
- Dixon, W.J. (1980). Efficient analysis of experimental observations. *Annu. Rev. Pharmacol. Toxicol.* 20, 441–462.
- Duan, B., Cheng, L., Bourane, S., Britz, O., Padilla, C., Garcia-Campany, L., Krashes, M., Knowlton, W., Velasquez, T., Ren, X., et al. (2014). Identification of spinal circuits transmitting and gating mechanical pain. *Cell* 159, 1417–1432.
- Duncan, G.E., Moy, S.S., Perez, A., Eddy, D.M., Zinzow, W.M., Lieberman, J.A., Snouwaert, J.N., and Koller, B.H. (2004). Deficits in sensorimotor gating and tests of social behavior in a genetic model of reduced NMDA receptor function. *Behav. Brain Res.* 153, 507–519.
- Eccles, J.C., Schmidt, R., and Willis, W.D. (1963a). Pharmacological studies on presynaptic inhibition. *J. Physiol.* 168, 500–530.
- Eccles, J.C., Schmidt, R.F., and Willis, W.D. (1963b). Depolarization of the central terminals of cutaneous afferent fibers. *J. Neurophysiol.* 26, 646–661.
- Ferguson, C., Hardy, S.L., Werner, D.F., Hileman, S.M., Delorey, T.M., and Homanics, G.E. (2007). New insight into the role of the $\beta 3$ subunit of the GABAA-R in development, behavior, body weight regulation, and anesthesia revealed by conditional gene knockout. *BMC Neurosci.* 8, 85.
- Fone, K.C.F., and Porkess, M.V. (2008). Behavioural and neurochemical effects of post-weaning social isolation in rodents—relevance to developmental neuropsychiatric disorders. *Neurosci. Biobehav. Rev.* 32, 1087–1102.
- Frank, K., and Fuortes, M.G.F. (1957). Presynaptic and postsynaptic inhibition of monosynaptic reflexes. *Fed. Proc.* 16, 39–40.
- Gödde, K., Gschwend, O., Puchkov, D., Pfeffer, C.K., Carleton, A., and Jentsch, T.J. (2016). Disruption of Kcc2-dependent inhibition of olfactory bulb output neurons suggests its importance in odour discrimination. *Nat. Commun.* 7, 12043.
- Grimes, W.N., Seal, R.P., Oesch, N., Edwards, R.H., and Diamond, J.S. (2011). Genetic targeting and physiological features of VGLUT3+ amacrine cells. *Vis. Neurosci.* 28, 381–392.
- Hasegawa, H., Abbott, S., Han, B.-X., Qi, Y., and Wang, F. (2007). Analyzing somatosensory axon projections with the sensory neuron-specific *Advillin* gene. *J. Neurosci.* 27, 14404–14414.
- Hooks, B.M., Lin, J.Y., Guo, C., and Svoboda, K. (2015). Dual-channel circuit mapping reveals sensorimotor convergence in the primary motor cortex. *J. Neurosci.* 35, 4418–4426.
- Iwashita, N., Nosaka, S., and Koyama, N. (2012). Involvement of peripheral NMDA receptor in melittin-induced thermographic flare. *Neurochem. Res.* 37, 2222–2228.
- Jänig, W., Schmidt, R.F., and Zimmermann, M. (1968). Two specific feedback pathways to the central afferent terminals of phasic and tonic mechanoreceptors. *Exp. Brain Res.* 6, 116–129.
- Jiménez, I., Rudomin, P., and Solodkin, M. (1987). Mechanisms involved in the depolarization of cutaneous afferents produced by segmental and descending inputs in the cat spinal cord. *Exp. Brain Res.* 69, 195–207.
- Kanaka, C., Ohno, K., Okabe, A., Kuriyama, K., Itoh, T., Fukuda, A., and Sato, K. (2001). The differential expression patterns of messenger RNAs encoding

- K-Cl cotransporters (KCC1,2) and Na-K-2Cl cotransporter (NKCC1) in the rat nervous system. *Neuroscience* 104, 933–946.
- Lau, J., Minett, M.S., Zhao, J., Dennehy, U., Wang, F., Wood, J.N., and Bogdanov, Y.D. (2011). Temporal control of gene deletion in sensory ganglia using a tamoxifen-inducible Advillin-Cre-ERT2 recombinase mouse. *Mol. Pain* 7, 100.
- Lee, C.J., Bardoni, R., Tong, C.-K., Engelman, H.S., Joseph, D.J., Magherini, P.C., and MacDermott, A.B. (2002). Functional expression of AMPA receptors on central terminals of rat dorsal root ganglion neurons and presynaptic inhibition of glutamate release. *Neuron* 35, 135–146.
- Li, J., McRoberts, J.A., Nie, J., Ennes, H.S., and Mayer, E.A. (2004). Electrophysiological characterization of N-methyl-D-aspartate receptors in rat dorsal root ganglia neurons. *Pain* 109, 443–452.
- Li, L., Rutlin, M., Abaira, V.E., Cassidy, C., Kus, L., Gong, S., Jankowski, M.P., Luo, W., Heintz, N., Koerber, H.R., et al. (2011). The functional organization of cutaneous low-threshold mechanosensory neurons. *Cell* 147, 1615–1627.
- Lidierth, M. (2006). Local and diffuse mechanisms of primary afferent depolarization and presynaptic inhibition in the rat spinal cord. *J. Physiol.* 576, 309–327.
- Liu, H., Wang, H., Sheng, M., Jan, L.Y., Jan, Y.N., and Basbaum, A.I. (1994). Evidence for presynaptic N-methyl-D-aspartate autoreceptors in the spinal cord dorsal horn. *Proc. Natl. Acad. Sci. USA* 91, 8383–8387.
- Luo, W., Enomoto, H., Rice, F.L., Milbrandt, J., and Ginty, D.D. (2009). Molecular identification of rapidly adapting mechanoreceptors and their developmental dependence on ret signaling. *Neuron* 64, 841–856.
- Malan, T.P., Mata, H.P., and Porreca, F. (2002). Spinal GABA(A) and GABA(B) receptor pharmacology in a rat model of neuropathic pain. *Anesthesiology* 96, 1161–1167.
- Marvión, J.C.G., McRoberts, J.A., Ennes, H.S., Song, B., Wang, X., Jinton, L., Corneliusen, B., and Mayer, E.A. (2002). Two N-methyl-D-aspartate receptors in rat dorsal root ganglia with different subunit composition and localization. *J. Comp. Neurol.* 446, 325–341.
- McRoberts, J.A., Ennes, H.S., Marvión, J.C.G., Fanselow, M.S., Mayer, E.A., and Vissel, B. (2011). Selective knockdown of NMDA receptors in primary afferent neurons decreases pain during phase 2 of the formalin test. *Neuroscience* 172, 474–482.
- Minett, M.S., Quick, K., and Wood, J.N. (2011). Behavioral measures of pain thresholds. *Curr. Protoc. Mouse Biol.* 1, 383–412.
- Nguyen, Q.-A., and Nicoll, R.A. (2018). The GABA_A receptor β subunit is required for inhibitory transmission. *Neuron* 98, 718–725.e3.
- Niederkofer, V., Asher, T.E., Okaty, B.W., Rood, B.D., Narayan, A., Hwa, L.S., Beck, S.G., Miczek, K.A., and Dymecki, S.M. (2016). Identification of serotonergic neuronal modules that affect aggressive behavior. *Cell Rep.* 17, 1934–1949.
- Orefice, L.L., Zimmerman, A.L., Chirila, A.M., Sleboda, S.J., Head, J.P., and Ginty, D.D. (2016). Peripheral mechanosensory neuron dysfunction underlies tactile and behavioral deficits in mouse models of ASDs. *Cell* 166, 299–313.
- Pagadala, P., Park, C.-K., Bang, S., Xu, Z.-Z., Xie, R.-G., Liu, T., Han, B.-X., Tracey, W.D., Jr., Wang, F., and Ji, R.-R. (2013). Loss of NR1 subunit of NMDARs in primary sensory neurons leads to hyperexcitability and pain hypersensitivity: involvement of Ca(2+)-activated small conductance potassium channels. *J. Neurosci.* 33, 13425–13430.
- Paul, J., Zeilhofer, H.U., and Fritschy, J.-M. (2012). Selective distribution of GABA(A) receptor subtypes in mouse spinal dorsal horn neurons and primary afferents. *J. Comp. Neurol.* 520, 3895–3911.
- Peirs, C., Williams, S.P., Zhao, X., Walsh, C.E., Gedeon, J.Y., Cagle, N.E., Goldring, A.C., Hioki, H., Liu, Z., Marell, P.S., and Seal, R.P. (2015). Dorsal horn circuits for persistent mechanical pain. *Neuron* 87, 797–812.
- Penzo, M.A., Robert, V., Tucciarone, J., De Bundel, D., Wang, M., Van Aelst, L., Darvas, M., Parada, L.F., Palmiter, R.D., He, M., et al. (2015). The paraventricular thalamus controls a central amygdala fear circuit. *Nature* 519, 455–459.
- Raccuglia, D., McCurdy, L.Y., Demir, M., Gorur-Shandilya, S., Kunst, M., Emonet, T., and Nitabach, M.N. (2016). Presynaptic GABA receptors mediate temporal contrast enhancement in Drosophila olfactory sensory neurons and modulate odor-driven behavioral kinetics. *eNeuro* 3, ENEURO.0080-16.2016.
- Rau, K.K., McIlwrath, S.L., Wang, H., Lawson, J.J., Jankowski, M.P., Zylka, M.J., Anderson, D.J., and Koerber, H.R. (2009). Mrgprd enhances excitability in specific populations of cutaneous murine polymodal nociceptors. *J. Neurosci.* 29, 8612–8619.
- Rudomin, P., and Schmidt, R.F. (1999). Presynaptic inhibition in the vertebrate spinal cord revisited. *Exp. Brain Res.* 129, 1–37.
- Russo, R.E., Delgado-Lezama, R., and Hounsgaard, J. (2000). Dorsal root potential produced by a TTX-insensitive micro-circuitry in the turtle spinal cord. *J. Physiol.* 528, 115–122.
- Rutlin, M., Ho, C.-Y., Abaira, V.E., Cassidy, C., Bai, L., Woodbury, C.J., and Ginty, D.D. (2014). The cellular and molecular basis of direction selectivity of A δ -LTMRs. *Cell* 159, 1640–1651.
- Shreckengost, J., Calvo, J., Quevedo, J., and Hochman, S. (2010). Bicuculline-sensitive primary afferent depolarization remains after greatly restricting synaptic transmission in the mammalian spinal cord. *J. Neurosci.* 30, 5283–5288.
- Stirling, L.C., Forlani, G., Baker, M.D., Wood, J.N., Matthews, E.A., Dickenson, A.H., and Nassar, M.A. (2005). Nociceptor-specific gene deletion using heterozygous Nav1.8-Cre recombinase mice. *Pain* 113, 27–36.
- Todd, A.J. (1996). GABA and glycine in synaptic glomeruli of the rat spinal dorsal horn. *Eur. J. Neurosci.* 8, 2492–2498.
- Torsney, C., and MacDermott, A.B. (2006). Disinhibition opens the gate to pathological pain signaling in superficial neurokinin 1 receptor-expressing neurons in rat spinal cord. *J. Neurosci.* 26, 1833–1843.
- Tsien, J.Z., Huerta, P.T., and Tonegawa, S. (1996). The essential role of hippocampal CA1 NMDA receptor-dependent synaptic plasticity in spatial memory. *Cell* 87, 1327–1338.
- Whitehorn, D., and Burgess, P.R. (1973). Changes in polarization of central branches of myelinated mechanoreceptor and nociceptor fibers during noxious and innocuous stimulation of the skin. *J. Neurophysiol.* 36, 226–237.
- Yan, X., Jiang, E., Gao, M., and Weng, H.R. (2013). Endogenous activation of presynaptic NMDA receptors enhances glutamate release from the primary afferents in the spinal dorsal horn in a rat model of neuropathic pain. *J. Physiol.* 591, 2001–2019.
- Zeng, J., Thomson, L.M., Aicher, S.A., and Terman, G.W. (2006). Primary afferent NMDA receptors increase dorsal horn excitation and mediate opiate tolerance in neonatal rats. *J. Neurosci.* 26, 12033–12042.
- Zhang, W., Liu, L.Y., and Xu, T.L. (2008). Reduced potassium-chloride cotransporter expression in spinal cord dorsal horn neurons contributes to inflammatory pain hypersensitivity in rats. *Neuroscience* 152, 502–510.

STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Guinea pig polyclonal anti-vGlut1	Millipore	Cat# AB5905; RRID: AB_2301751
Mouse monoclonal anti-NR1 (clone 54.1)	Millipore	Cat# MAB363; RRID: AB_94946
Chicken polyclonal anti-GFP	Aves	Cat# GFP-1020; RRID: AB_10000240
Rabbit polyclonal anti-NF200	Sigma	Cat# N4142; RRID: AB_477272
Rabbit polyclonal anti-GABRB3	Biomatik, this paper	N/A
Isolectin B4	LifeTech	Cat# I32450; RRID: SCR_014365
Bacterial and Virus Strains		
AAV2/9- fDIO-Cre-GFP	Boston Children's viral core; Penzo et al., 2015	N/A
Chemicals, Peptides, and Recombinant Proteins		
Gabazine (SR95531)	Sigma-Aldrich	Cat# S106
Bicuculline methochloride	Tocris	Cat# 0130
Strychnine	Sigma-Aldrich	Cat# S0532
2,3-Dioxo-6-nitro-1,2,3,4-tetrahydrobenzo[f]quinoxaline-7-sulfonamide (NBQX)	Fisher Scientific	Cat# 037310
6-Cyano-7-nitroquinoxaline-2,3-dione (CNQX)	Sigma-Aldrich	Cat# C127
(RS)-3-(2-Carboxypiperazin-4-yl)-propyl-1-phosphonic acid (CPP)	Fisher Scientific	Cat# 017310
(α S)- α -Amino-3-[(4-carboxyphenyl)methyl]-3,4-dihydro-5-iodo-2,4-dioxo-1(2H)-pyrimidinepropanoic acid (UBP301)	Tocris	Cat# 1766
Tetrodotoxin citrate (TTX)	Tocris	Cat# 1069
4-aminopyridine (4-AP)	Tocris	Cat# 0940
Tubocurarine chloride pentahydrate	Sigma-Aldrich	Cat# 93750
Mecamylamine hydrochloride	Sigma-Aldrich	Cat# M9020
(2S)-(+)-5,5-Dimethyl-2-morpholineacetic acid (SCH 50911)	Tocris	Cat# 0984
(2S)-2-Amino-2-[(1S,2S)-2-carboxycycloprop-1-yl]-3-(xanth-9-yl)propanoic acid disodium salt (LY 341495)	Tocris	Cat# 4062
(RS)- α -Methyl-4-carboxyphenylglycine disodium salt (MCPG)	Tocris	Cat# 3696
Critical Commercial Assays		
SsoAdvanced Universal SYBR Green Supermix	Bio-Rad	Cat# 1725271
SuperScript IV VILO Master Mix with ezDNase Enzyme	Thermo Fisher	Cat# 11766050
Direct-zol RNA MicroPrep	Fisher (Zymo Research)	Cat# R2060
RNAscope Fluorescent Multiplex Assay	ACD Bio	Cat# 320850
Experimental Models: Organisms/Strains		
Mouse: <i>Grin1^{fl/fl}</i> ; B6.129S4- <i>Grin1^{tm2Stl/J}</i>	The Jackson Laboratory	RRID: IMSR_JAX:005246
Mouse: <i>Gabrb3^{fl/fl}</i> ; B6;129- <i>Gabrb3^{tm2.1Geh/J}</i>	The Jackson Laboratory	RRID: IMSR_JAX:008310
Mouse: <i>Nav1.8^{Cre}</i>	Stirling et al., 2005	RRID: IMSR_EM:04582
Mouse: <i>Advillin^{Cre}</i>	Hasegawa et al., 2007	RRID: IMSR_JAX:032536
Mouse: <i>Advillin^{CreER}</i> ; B6.Cg-Tg(Avil-icre/ERT2)AJwo/J	The Jackson Laboratory	RRID: IMSR_JAX:026516
Mouse: <i>Ret^{CreER}</i>	Luo et al., 2009	RRID: MGI:4437245
Mouse: <i>Mrgd^{Cre}</i>	Rau et al., 2009	RRID: MGI:3852409
Mouse: <i>Lbx1^{flpo}</i>	Duan et al., 2014	RRID: MGI:5770780
Mouse: Ai32; B6.Cg-Gt(ROSA) ^{26Sortm32(CAG-COP4*H134R/EYFP)Hze/J}	The Jackson Laboratory	RRID: IMSR_JAX:012569
Mouse: <i>Advillin^{flpo}</i>	to be characterized in a subsequent paper	MGI: 1333798

(Continued on next page)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Mouse: TrkB ^{CreER}	Rutlin et al., 2014	RRID: IMSR_JAX:027214
Mouse: CGRP ^{Flpe}	to be characterized in a subsequent paper	MGI: 2151253
Mouse: TH ^{2A-CreER} ; Th ^{tm1.1(cre/ERT2)Ddg/J}	The Jackson Laboratory	RRID: IMSR_JAX:025614
Mouse: Rosa26 ^{LSL-FSF-ReaChR} ; B6;129S-Gt(ROSA)26Sor ^{tm2.1Ksv0/J}	The Jackson Laboratory	RRID: IMSR_JAX:024846
Mouse: Rosa26 ^{FSF-ReaChR}	The Jackson Laboratory	Derived from Rosa26 ^{LSL-FSF-ReaChR}
Mouse: Rosa26 ^{LSL-ReaChR}	The Jackson Laboratory	Derived from Rosa26 ^{LSL-FSF-ReaChR}
Mouse: Vglut3 ^{Cre} ; Tg(Slc17a8-icre)1Edw/SealJ	The Jackson Laboratory	RRID: IMSR_JAX:018147
Mouse: CCK ^{Cre} ; Cck ^{tm1.1(cre)Zjh/J}	The Jackson Laboratory	RRID: IMSR_JAX:012706
Oligonucleotides		
GAPDH qPCR primers forward: ACCCTGTTGCTGTAGCCGTATCA reverse: TCAACAGCAACTCCCACTCTCCA	Chen et al., 2014	N/A
NR1 qPCR primers forward: CCTGGAAGCAAAATGTGTCCC reverse: AGATCCCAGCTACGATGCCT	This paper	N/A
GABRB3 qPCR primers forward: GCCAGCATCGACATGGTTTC reverse: GCGGATCATGCGGTTTTTCA	Chen et al., 2014	N/A
RNAscope Probe- Mm-Grin1-O2	ACD Bio	Cat# 474011
RNAscope Probe- Mm-Nefh	ACD Bio	Cat# 443671
RNAscope Probe- Mm-Mrgprd	ACD Bio	Cat# 417921
RNAscope Probe- Mm-Slc17a8	ACD Bio	Cat# 431261
RNAscope Probe- EGFP	ACD Bio	Cat# 400281
RNAscope Probe- CRE	ACD Bio	Cat# 312281

CONTACT FOR REAGENT AND RESOURCE SHARING

Further information and requests for resources and reagents should be directed to the Lead Contact, David Ginty (david_ginty@hms.harvard.edu).

EXPERIMENTAL MODEL AND SUBJECT DETAILS

Animals

Mice for behavior were ear notched and tail biopsied at postnatal day 21 and weaned at postnatal day 25. All behavioral experiments were completed between 8-11 weeks of age. Mice for electrophysiology were toe tagged at postnatal day 5-7 and used at postnatal day 14-18. In every behavioral and physiology experiment where mutants were compared to control mice, the experimenter was blinded to genotype. All experiments were in accordance with protocols approved by Harvard Medical School IACUC. The following mouse lines were used in electrophysiology and behavioral experiments:

Grin1^{fl/fl} mice were obtained from Jackson Laboratory (RRID: IMSR_JAX:005246) and were previously described ([Tsien et al., 1996](#)). *Gabrb3^{fl/fl}* mice were obtained from Jackson Laboratory (RRID: IMSR_JAX:008310) and were previously described ([Ferguson et al., 2007](#)). Mice were generated by crossing a double heterozygous (Cre/+, f/+) male to homozygous conditional females. Cre negative littermates were used as controls. The following Cre and flp lines were used for conditional knockout strategies: *Avil^{Cre}* (RRID: IMSR_JAX:032536; [Hasegawa et al., 2007](#)), *Avil^{CreER}* (RRID: IMSR_JAX:026516; [Lau et al., 2011](#)), *Nav1.8^{Cre}* (RRID: IMSR_EM:04582; [Stirling et al., 2005](#)), and *Avil^{Flpo}* (to be characterized elsewhere, T. Dickendesher and D.D.G., unpublished data). Tamoxifen was dissolved in sunflower oil (10mg/mL) and stored at -20 in 1mL aliquots, and administered to *Avil^{CreER}*, *Grin1^{fl/fl}* and *Avil^{CreER}*; *Gabrb3^{fl/fl}* mice and their control littermates by intraperitoneal injections for 1 mg per day for 5 days. *Grin1^{fl/fl}* and *Gabrb3^{fl/fl}* mice used for behavioral experiments were on mixed genetic backgrounds (C57Bl6/CD1 for *Grin1* and Sv129/C57Bl6 for *Gabrb3*).

For optogenetic physiology experiments, the following Cre and Flp lines were used: *Nav1.8^{Cre}* (RRID: IMSR_EM:04582; [Stirling et al., 2005](#)), *TrkB^{CreER}* (tamoxifen treated at p7 or p8, 0.5mg; RRID: IMSR_JAX:027214; [Rutlin et al., 2014](#)); *CGRP^{Flpe}* (to be characterized in a subsequent paper); *Ret^{CreER}* (tamoxifen treated 2.5mg at e10.5; RRID: MGI:4437245; [Luo et al., 2009](#)); *TH^{2A-CreER}* (tamoxifen treated at p8 or p10, 0.5mg, RRID: IMSR_JAX:025614; [Abraira et al., 2017](#)), *Mrgd^{Cre}* (RRID: MGI:3852409; [Rau et al., 2009](#)), *Vglut3^{Cre}* (Jackson laboratory, RRID: IMSR_JAX:018147; [Grimes et al., 2011](#)), *Lbx1^{Flpo}* (RRID: MGI:5770780; [Duan et al., 2014](#)), *CCK^{Cre}* (Jackson Laboratory, RRID: IMSR_JAX:012706; [Abraira et al., 2017](#)). Conditional optogenetic lines: Rosa26^{LSL-ChR2} (Ai32, RRID: IMSR_JAX:012569); Rosa26^{LSL-FSF-ReaChR}; Rosa26^{FSF-ReaChR} (RRID: IMSR_JAX:024846; [Hooks et al., 2015](#)).

METHOD DETAILS

Electrophysiology

In vitro DRP measurements were completed as described previously (Orefice et al., 2016). Postnatal day 14–18 mice of both sexes were anesthetized with isofluorane, transcardiacally perfused with ice cold high Mg^{2+} , low Ca^{2+} ACSF containing: 128mM NaCl, 1.9mM KCl, 1.2mM KH_2PO_4 , 26mM $NaHCO_3$, 0.85mM $CaCl_2$, 6.5mM $MgSO_4$, and 10mM d-glucose (pH of 7.4) prior to decapitation and vertebral column removal. The vertebral column was placed in a circulating, oxygenated bath of high Mg, low Ca ACSF, and a ventral vertebrectomy and laminectomy was performed. An off-centered sagittal hemisection (adjacent to the dorsal column) was performed with insect pins, then the preparation was inverted and dorsal laminectomy completed. The isolated cord was secured with insect pins in a sylgard covered dish, and dura around the spinal cord and roots removed to allow suction electrode access. The superfusate was then switched to recording ACSF (128mM NaCl, 1.9mM KCl, 1.2mM KH_2PO_4 , 26mM $NaHCO_3$, 2.4mM $CaCl_2$, 1.3mM $MgSO_4$, and 10mM D-glucose) circulating (3–5mL/minute).

A L1–T10 dorsal root was stimulated, with a recording suction electrode placed proximally on the same root to record afferent volleys. An additional suction electrode was placed on the rostral adjacent dorsal root close to its site of entry into the spinal cord. Stimulus threshold was determined by gradually increasing stimulus intensity until an afferent volley was detected in at least 50% of the sweeps (typically 3–6 μ A; stimulus duration 100 μ s). DRPs were recorded at 2T, 3T, and at 500 μ A, 500 μ s for maximal activation, for a total of 10 sweeps each, with 10 s between sweeps for 2T and 3T and 30 s between sweeps for 500 μ A, 500 μ s.

For dorsal cutaneous nerve activation, the preparation was similar as above, but back hairy skin was shaved prior to perfusion and removed with the vertebral column. After ventral vertebrectomy, laminectomy, and hemisection were completed, back hairy skin was pinned down and cut along the midline, minimizing disruption of the main branches of the dorsal cutaneous nerves. The right side of the vertebral column and associated dorsal cutaneous nerves were kept intact during the dorsal vertebrectomy, and fascia partially removed from the dorsal cutaneous nerves to allow a stimulating suction electrode access. Afferent volleys were recorded in the corresponding dorsal root, and DRPs from the rostrally located adjacent root. All ventral roots were cut to remove motor signals to the intercostal muscles.

DC recordings were acquired using an AC/DC differential amplifier and mini headstage (A-M systems, Model 3000), a Digidata 1440A acquisition system, and pClamp10 software (Molecular Devices), using a 10kHz sampling rate. A custom-built MATLAB program was used to subtract the baseline values prior to the stimulus, average 10 sweeps at each intensity, low-pass filter the response at 100 Hz, find the onset of the evoked DRP, its peak amplitude, the integral under the filtered response from onset peak and from peak to offset (defined as the time at which the slow potential decayed to 20% of peak amplitude, with a 1 s hard time cap). When comparing integral changes with the addition of drug, parameters were constrained to baseline values. Traces shown are average of 10 sweeps and low-pass filtered at 1 kHz. Stimulus artifact and occasionally large afferent volleys were truncated manually offline for visualization purposes.

To activate ChR2 or ReaChR in the isolated spinal cord, whole field illumination was applied through a 10x objective centered over the dorsal root entry zone. Afferent terminals or interneurons were stimulated with blue or yellow light (473 and 555nm, 30–50mW at the level of the spinal cord, respectively, for 10ms). When comparing baseline values for conditional knockout mice, mutants, and their littermate controls lacking Cre were recorded on the same day when possible or on adjacent days, with the experimenter blinded to genotype.

For any drug treatment experiments, stable recordings were obtained for at least 15 min before adding drug, and when possible without DC fluctuations in resting potential, washout recordings obtained. The following drugs were added to the circulating ACSF of known volume to make the noted final concentration: SR95531 (gabazine, 5–10 μ M, Sigma-Aldrich, S106), bicuculline methochloride (10–20 μ M, Tocris, 0130), strychnine (1 μ M, Sigma Aldrich, S0532), NBQX (2,3-Dioxo-6-nitro-1,2,3,4-tetrahydrobenzo[f]quinoxaline-7-sulfonamide, 10 μ M, Fisher Scientific, 037310), CNQX (6-Cyano-7-nitroquinoxaline-2,3-dione, 10 μ M, Sigma-Aldrich, C127), CPP ((RS)-3-(2-Carboxypiperazin-4-yl)-propyl-1-phosphonic acid, 20 μ M, Fisher Scientific 017310), UBP301 ((α S)- α -Amino-3-[(4-carboxyphenyl)methyl]-3,4-dihydro-5-iodo-2,4-dioxo-1(2H)-pyrimidinepropanoic acid, 10 μ M, Tocris, 1766), tetrodotoxin citrate (TTX, 1 μ M, Tocris, 1069), 4-aminopyridine (4-AP, 400 μ M–1mM, Tocris, 0940), tubocurarine chloride pentahydrate (20 μ M, Sigma Aldrich, 93750), mecamyline hydrochloride (50 μ M, Sigma-Aldrich, M9020), SCH 50911 ((2S)-(+)-5,5-Dimethyl-2-morpholineacetic acid, 10 μ M, Tocris, 0984), LY341495 disodium salt ((2S)-2-Amino-2-[(1S,2S)-2-carboxycycloprop-1-yl]-3-(xanth-9-yl) propanoic acid, 100 μ M, Tocris, 4062), and (RS)-MCPG disodium salt ((RS)- α -Methyl-4-carboxyphenylglycine, 100 μ M, Tocris, 3696).

For gabazine subtraction experiments (Figure 2E), average traces of 10 sweeps from 500 μ A, 500 μ s stimulation were low-pass filtered at 100Hz and drug induced traces subtracted from baseline traces. Peak time, peak amplitude, trough time, and trough amplitude were determined using PClamp and expressed as a % of baseline amplitude.

For recruitment threshold comparisons, within the same two-week period, spinal cords and dorsal roots were isolated from P14–P18 mice or 4–6 month Cre negative behavior mice treated with tamoxifen. A sacral root was stimulated distally and recorded at the DR entry zone in the presence of ionotropic glutamate receptor antagonists CPP and NBQX. This recording configuration allowed at least 3.5mm from stimulation to recording site in order to clearly distinguish slower conducting afferent volleys.

Tissue Fixation and Immunohistochemistry

Male and female mice 6–52 weeks old and their age and sex matched littermate controls were anesthetized with isoflurane and perfused with 5–10 mL modified Ames Media in 1x PBS, followed by 0.5 mL/g of 2% PFA (for NR1 or GABRB3 IHC) or 4% PFA (for all other IHC experiments), with no postfix. Brains, spinal cords, and DRGs were dissected out and kept in PBS at 4°C. For long-term storage, tissue was stored in PBS + 0.01% sodium azide.

Tissue for vibratome sectioning (only used in the *Vglut3^{Cre}*, *Lbx1^{Flpo}*, *Rosa26^{LSL-FSF-synaptophysin-GFP}* and *CCK^{Cre}*, *Rosa26^{LSL-ChR2}* crosses) was mounted on 4% agar in desired orientation and covered with 2% agar to hold it in place. 60 µm sagittal sections of spinal cord were made on the vibratome, placed in wells filled with 1x PBS, and stored at 4°C.

Sagittal sections were processed for free floating immunohistochemistry as described previously (Abraira et al., 2017). Tissue sections were incubated in 50% ethanol in ddH₂O for 30 min to improve antibody penetration, then washed in high salt 1x PBS (1x PBS with 0.3M NaCl, HS-PBS) 3x10 min. Sections were then incubated with primary antibodies diluted in HS-PBS with 0.3% Triton X-100 (HS-PBST) for 48–72 h at 4°C. Sections were washed in HS-PBST 3x10 min, and then incubated with secondary antibodies diluted in HS-PBST overnight at 4°C. Sections were washed in HS-PBST 3x10 min, and then mounted on glass slides and coverslipped with Fluoromount Aqueous Mounting Medium (Sigma). The slides were stored at 4°C.

Tissue for frozen sectioning (all other genetic crosses) was cryoprotected in 30% sucrose in 1x PBS overnight at 4°C, embedded in OCT and frozen in tissue molds over a dry ice and ethanol slurry, then stored at –80°C until sectioning. 20 µm transverse sections of thoracic spinal cord and DRGs were made on a cryostat, with littermate controls and mutants on the same slides, and stored at –80°C until staining.

Sections were defrosted for 1 h at room temperature, then rehydrated for 3x10 min in 1x PBS. Slides were blocked for 1–2 h in 1x PBS containing 0.1% Triton X-100 and 5% normal serum (Vector Labs S-1000). Sections were incubated with primary antibodies diluted in blocking solution (5% donkey or goat serum, no detergent) overnight at 4°C. Slides were washed 4x5 min in 1x PBS containing 0.02% Tween 20 (PBSt) and then incubated with species specific secondary antibodies in blocking solution for 1 h at room temperature. Tissue was washed with PBSt 2x5 min, then treated with 4',6-diamidino-2-phenylindole (DAPI) stain at a 1:5000 dilution for 10 min, then washed again with PBSt for 2x5 min. Tissue sections were then mounted on glass slides and coverslipped with Fluoromount Aqueous Mounting Medium (Sigma). The slides were stored at 4°C.

For GABRB3 staining and NR1 staining age comparisons, a modified high salt protocol was used on frozen 20 µm sections. Sections were defrosted for 5–12 h at room temperature, then rehydrated for 3x10 min in 1x PBS. Tissue sections were incubated in 50% ethanol in ddH₂O for 30 min to improve antibody penetration, then washed in HS-PBS for 3x10 min. Sections were then incubated at 37°C for 30 min in HS-PBS, then for 15 min in 0.001% trypsin + 0.001% CaCl₂ at 37°C. Slides were washed in HS-PBS 3x10 min. Sections were incubated with primary antibodies diluted in HS-PBST for 48 h at 4°C. After 48 h, sections were incubated with primary antibodies for 30 min at room temperature. Sections were then washed in HS-PBST 6x10 min, and then incubated with secondary antibodies diluted in HS-PBST for 2 h at room temperature. Sections were washed in HS-PBST 3x10 min, and then mounted on glass slides with Fluoromount Aqueous Mounting Medium (Sigma). The slides were stored at 4°C.

The following primary antibodies and concentrations were used: guinea pig anti-VGlu1 (Millipore AB5905, 1:1000, RRID: AB_2301751); mouse anti-NR1 (Millipore MAB 363, 1:500, RRID: AB_94946), chicken anti-GFP (Aves, GFP-1020, 1:1000, RRID: AB_10000240), IB4-647 (LifeTech I32450, 1:500, RRID: SCR_014365), rabbit anti-NF200 (Sigma N4142, 1:1000, RRID: AB_477272). A rabbit polyclonal antibody to GABRB3 was raised against the full-length protein, with the signal peptide and transmembrane domains deleted. The final sequence represented residues 26–245, 328–450 of mouse GABRB3 (Biomatik, Ontario, Canada, 1:500). An array of goat derived Alexa Fluor 488,546, and 647 secondaries conjugated IgGs (Invitrogen) were used.

RT-qPCR

Male and female mice 11–29 weeks old and their age and sex matched littermate *Grin1^{f/+}* or *Gabrb3^{f/+}* controls were sacrificed by asphyxiation followed by cervical dislocation and decapitation. Dissections were kept to less than 25 min from time of death and occurred on a lab bench that had been thoroughly cleaned with 70% ethanol and ELIMINase (VWR). DRGs were dissected out and placed immediately into RNAlater solution (Invitrogen) on ice. DRGs in RNAlater solution were stored at 4°C.

To extract RNA, DRGs were removed from RNAlater solution and placed into 300 µL of TRIzol (Thermo Fisher). DRGs were disrupted in TRIzol using a motorized hand-held pestle and homogenized with a 31-gauge needle.

Direct-zol Microprep Purification Column (Zymo Research) was used to extract RNA from homogenized samples, with a DNase I treatment step on-column. Extracted RNA samples were stored at –80°C. Reverse transcription was completed with the SuperScript IV VILO Master Mix after treatment with ezDNase enzyme (Thermo Fisher), using 100 ng of RNA per sample.

Samples were assayed using SsoAdvanced Universal SYBR Green Supermix and run on a CT1000 Thermal Cycler with CFX96 Real-Time System (BioRad). cDNA was diluted by a factor of 10, and 5 µL of sample cDNA was used per 20 µL reaction. Cycle conditions were: 95°C for 30 s, then 40 cycles of 95°C for 10 s and 60°C for 30 s, ending with a melt curve analysis ranging from 65°C to 95°C with 5 s steps of 0.5°C increments. All reactions were run in triplicate, with individual wells excluded from analysis if they differed beyond ± 0.2 quantification cycles (Cq) from the remaining two wells. RNA expression levels were determined from the mean values, and were standardized to expression of the housekeeping gene glyceraldehyde 3-phosphate dehydrogenase (GAPDH). Data shown is normalized to average values from control mice. The following primer pairs were used: GAPDH (forward): ACCCTGTTGC TGTAGCCGTATCA and (reverse): TCAACAGCAACTCCCACTCTCCA; GABRB3 (forward): GCCAGCATCGACATGGTTTC and

(reverse): GCGGATCATGCGGTTTTTCA; NR1 (forward): CCTGGAAGCAAAATGTGTCCC and (reverse): AGATCCCAGCTAC GATGCCT.

RNAscope

Male and female mice 13–41 weeks old and their age and sex matched littermate controls were sacrificed by asphyxiation followed by cervical dislocation and decapitation. Dissections were kept to less than 25 min from time of death and occurred on a sterilized lab bench that had been thoroughly cleaned with 70% ethanol and ELIMINase (VWR). Thoracic DRGs and spinal cords were dissected out and placed into a cryomold filled with OCT, which was then placed in a dry-ice cooled methyl-butane bath until frozen. Frozen samples were stored in airtight containers at -80°C .

RNAscope staining was completed following the RNAscope Fluorescent Multiplex Assay (ACD Bio). The following probes were used: Mm-Grin1-O2-C3 (ACDBio, 474011-C3), Mm-Nefh (ACDBio, 443671), Mm-Mrgprd-C2 (ACDBio, 417921-C2), Cre (ACDBio, 312281), Mm-SLC17a8 (ACDBio, 431261), and EGFP-C2 (ACDBio, 400281-C2). For combination with immunohistochemistry, the above protocol was run through until the hybridization of Amp 4-FL. After washing to remove Amp 4-FL, immunohistochemistry was completed following the protocol outlined above before counterstaining with DAPI.

Image analysis

Z stack images for puncta analysis were obtained on the Zeiss LSM7000 confocal microscope with the 63x objective. Images were taken in lamina III–IV, within $150\mu\text{m}$ of lamina II_{IV} (using IB4 labeling to demarcate boundary). The depth of the z stack was determined by the expanse of VGlut1⁺ terminals. Imaging parameters were held constant for NR1/GABRB3 across all animals. NR1 and GABRB3 puncta analysis was performed using a custom script in ImageJ over $0.9\mu\text{m}$ thick images, creating a mask of VGlut1⁺ terminals $\geq 0.5\mu\text{m}$ in diameter, thresholding NR1/GABRB3, and counting puncta of at least $0.1\mu\text{m}$ in diameter contained within that mask. 3–4 animals of each genotype or age were used, with at least 1000 VGlut1 terminals per animal.

For *CCK^{iCre}*, *Rosa26^{LSL-ChR2}* mouse RNAscope analysis, Z stack images were obtained on the Zeiss LSM7000 confocal microscope with the 20x objective. Imaging parameters were held constant for all samples, with a minimum of 6 dorsal horn images per animal. Images were analyzed by hand using the threshold of 5 distinct puncta within a DAPI positive cell to consider the cell positive for expression of the probe RNA.

Dorsal Column Injections

Male and female mice P28–P30 were anesthetized by isoflurane inhalation (1.5%–2.5%). Breathing rate and anesthesia were continuously monitored and isoflurane level adjusted as necessary. The back of the neck was shaved and then swabbed with betadine and 70% ethanol. A 5mm incision was made in the skin, and 0.5% lidocaine was applied to the incision site and underlying muscle. Muscles were separated or cut to expose the cervical vertebral column. The dura and arachnoid membranes between C1 and C2 were cut to expose the spinal cord. Four 75nL injections of AAV2/9- fDIO-Cre-GFP virus (Boston Children's viral core; $1.16492\text{E}+13$ gc/ml) mixed with a nominal volume of fast green were made bilaterally into the dorsal column under visual guidance with a glass pipette. Injections were assessed by determining the extent to which the dorsal column took up the tracer, and an additional injection site was added when inefficient uptake noted. Muscle and skin were then stitched together with sutures, and carprofen (4mg/kg) was injected intraperitoneally for analgesia. Mice were returned to their home cage after recovery from anesthesia, and additional doses of carprofen were given twice daily for 48 h post-surgery. The condition of the mice was monitored daily for 1 week following surgery, and weekly thereafter. Mice were used 4 weeks later for behavior or tissue processing.

Behavioral Assays

All mice were group housed with same-sex littermates with a 12-hour light/dark cycle and free access to food and water. Cages were changed once per week by an investigator at least three days before the next behavioral assay. Mice were habituated to behavioral test location while still in their home cage for a minimum of 30 min before testing began. All testing materials were cleaned thoroughly with ddH₂O and/or 70% ethanol before and between trials. All investigators were blind to animals' genotypes while running tests and gathering data.

Elevated Plus Maze (EPM)

The elevated plus maze was run in an acrylic chamber with arms 30 cm long and 5 cm wide. Two opposing arms have walls that were 15 cm high (closed arms), while the other two arms have no walls (open arms). The maze was elevated 40 cm above the ground, with a barrier around the base. Mice were allowed to freely explore the maze for 10 min under dim lighting. Video was recorded from overhead to track the mouse's movements. Videos were analyzed for the time spent in the open arms of the maze compared to time in closed arms or center using a custom MATLAB script.

Open Field Test

Open field test is a measure of anxiety-like behavior in rodents. Mice were placed individually into a 40 cm x 40 cm x 40 cm matte black acrylic testing chamber with matte white acrylic floor under dim lighting. Exploratory behavior was recorded from above and videos were tracked using custom MATLAB scripts to analyze distance traveled and time spent within a 5cm radius of the center of the chamber.

Novel Object Recognition Test (NORT)

Texture NORT was used as a measure of texture discrimination, as previously described (Orefice et al., 2016). For these experiments, we instituted specific handling protocols to ensure minimal stress during the repeated transfers between the home cage and testing chamber. Kimwipes were added to the wire cage hopper to encourage the mice to nest build and socialize, minimizing anxiety and male-to-male aggression. These kimwipes were used to scoop mice into the palm of the investigator's hand, applying gentle pressure to the tail to secure them during transfers, and mice were allowed to step off the investigator's hand at the end of the transfer.

For two consecutive days (day 1 and 2) mice were individually habituated to an empty testing chamber (40 cm x 40 cm x 40 cm) under dim lighting for 10 min. Mice were then tested on color/shape NORT (day 3) and texture NORT (day 4).

NORT has a learning phase and a testing phase. During the learning phase, the mouse was placed in the testing chamber with two identical objects that are placed equidistant from each other and the chamber walls. The mouse was allowed to freely explore the chamber and objects for 10 min. The mouse was then returned to its home cage for a 5 min retention period, during which the chamber and objects were thoroughly cleaned with 70% ethanol and one of the objects was replaced with a novel object. After the 5 min, the mouse was placed back in the chamber for the 10 min exploration of the testing phase.

Both the learning and testing phases were video-recorded from above. Custom MATLAB scripts were used to track mouse position in the chamber, and the amount of time the mouse spent investigating the objects in both the learning and testing phases was calculated. For color/shape NORT, the objects were wooden blocks that differ in shape and color. In texture NORT, the objects were plexiglass cubes (4 cm³) that were visually identical but varied in texture (rough or smooth). Mice were whisker plucked three days before the start of habituation. The preference for the novel object was calculated based solely on the time exploring either object.

Prepulse Inhibition (PPI) Startle Assay

The PPI Startle Assay was done using a San Diego Instruments startle reflex system (SR-LAB Startle Response System), as described (Orefice et al., 2016). Mice were tested individually in a cylindrical chamber within a soundproof chamber. A prepulse was delivered prior to an acoustic startle pulse, and the mouse's startle reflex was measured using an accelerometer mounted to the cylindrical holding chamber.

For tactile PPI the prepulse was an air puff administered to the back hairy skin. The air puff was a constant intensity (0.9PSI), but was presented at varying intervals before the startle pulse, ranging from 50ms to 1 s. For acoustic PPI the prepulse was a 20ms acoustic tone. The tone was always presented 100ms before the startle, but the decibel level varied across trials, ranging from 3dB to 15dB above background noise. Acoustic PPI was done with background noise at 65dB, while tactile PPI had background noise at 75dB to ensure the air puff prepulse could not be heard.

Von Frey Assay

Mice were placed into individual clear acrylic chambers, 4.5in x 2in x 2in, on a wire grid elevated to 14.5 in. Mice were habituated to the chambers for 60 min on two consecutive days and for 45 min before testing on the third day. Von Frey filaments were applied perpendicular to the plantar surface of a mouse's hind paw until the filament buckled, indicating its preset force was being applied to paw. For all crosses except those receiving dorsal column injections for *Gabrb3^{fl/fl}*, the repeated-measures Von Frey method was used (Minett et al., 2011). Each filament was used for 10 trials (pokes), with number of responses out of 10 recorded. A response was defined as withdrawing, licking, or shaking the paw in response to filament's application or immediately after. Filaments were used in order of increasing force, completing all 10 trials with a filament before moving on to the next. For DCN-injected *Gabrb3^{fl/fl}* mice and their littermate controls, the 'up and down' Von Frey method (Chaplan et al., 1994) was done to determine the 50% withdrawal thresholds, using the statistical method of Dixon (Dixon, 1980).

Textured Floor Aversion Assay

The modified handling protocol described above (see NORT) was used to minimize anxiety and limit aggression between mice. Tails were marked using a blue permanent soy ink marker for easy identification during transfers between the cage and testing chamber, and to further acclimate the mice to investigator handling. To mark tails mice were lifted gently from the home cage and placed on the wire cage hopper facing away from the investigator, held in place by gentle pressure on their tail.

Testing chambers were matte black acrylic boxes, 12in x 11.8in x 6in, placed on a matte white acrylic floor. For two consecutive days mice were habituated to the empty testing chamber under dim lighting. On the first day of habituation, mice were placed individually in the testing chamber and allowed to explore freely for 10 min before being returned to their home cage. On the second day

of habituation, mice underwent sociability habituation in which cagemates were habituated to the testing chamber together to decrease anxiety. If mice displayed stress in the chamber and/or during handling they were subjected to a second sociability habituation in a clean chamber to allow additional handling time.

The textured floor aversion assay is composed of an explore phase and a test phase. In the explore phase, the floor of the testing chamber is covered with a new sheet of pale pink control paper (construction). Mice were placed into the testing chamber individually and allowed to freely explore for 10 min. Mice were returned to their home cage while the testing chambers were thoroughly cleaned with 70% ethanol and reset for the test phase.

For the test phase, two sandpaper textures, SP400 (fine grit) and SP60 (coarse grit), were taped together on the back of the paper with a ½ inch overlap. White heavyweight construction paper was used to cover and protect the surface of the acrylic floor, and the sandpaper sheets were then placed over the construction paper. The test chambers were placed onto the sandpaper floors so that the seam of the joined sandpaper sheets lay in the middle of the test chamber, ensuring each grit covered half of the testing chamber. Mice were once again transferred from their home cage to the test chamber and allowed to explore the sandpaper textures for 10 min.

Both exploration and test phases were video-recorded from above. Custom MATLAB scripts were used to track mouse position in the chamber, calculating time spent on each half of the chamber and preference for sides of the chamber.

Estrus Testing (Vaginal Cytology) and Scoring

Estrus stage was determined using the vaginal lavage method. An individual female mouse was removed from the home cage and placed on the wire cage hopper facing away from investigator. Investigator grasped the mouse's tail to elevate its hind paws, allowing the mouse to hold the grid with its forepaws. A pipet with 80uL of sterile saline was used to gently flush the opening of the vaginal canal, avoiding penetration. Saline was expelled and withdrawn several times to ensure a sample, and then placed on a labeled glass slide, pre-lined with a hydrophobic barrier. Samples were left to dry at room temperature.

During a mouse's first sample an estrus response score was given based on the mouse's response to the pipet tip/saline administration. Any behavioral responses to handling were not included in the estrus score. Scoring was done by an investigator blinded to genotype as follows:

- 0 = passive, no resistance while being held, remains still as saline is administered
- 1 = pulling away from saline/pipette tip
- 2 = pulling away from saline/pipette tip, abdominal twisting to escape
- 3 = pulling away from saline/pipette tip, abdominal twisting to escape, kicking and/or turning to bite tip/ investigator's hands

Hargreaves Test

Mice were habituated for 45 min to individual chambers placed upon an elevated glass surface for two days before testing. On the day of testing, after a 30 min habituation period a radiant heat source was moved beneath glass surface and aimed at the hind paw of an animal. Upon withdrawal of the paw from the heat source, radiant heat delivery was terminated and the apparatus recorded the withdrawal latency, with a maximum stimulation of 20 s. Four rounds of testing were done, with 10 min between each radiant heat stimulus application and alternating paws to ensure a minimum of 20 min between stimulation of same paw. Withdrawal latencies across the four trials were averaged per animal.

Formalin Sensitivity Test

Mice were habituated to testing chambers, 8.68in x 4.65in x 5.5in, for an hour on two consecutive days. Mice were habituated in the chambers for 30 min before beginning the test, then removed individually from their chambers and anesthetized with isoflurane. Each mouse received a 20uL intradermal injection of 2% formalin in the right hind paw. Slight pressure was applied to the injection site to prevent backflow, and the mouse's respiratory rate was observed for 15-30 s before being placed back into the testing chamber.

Mice were recorded for one-hour post-injection. Videos were hand-scored for time spent licking and lifting the injected paw. All mice were sacrificed 2 h post-test.

Statistical Analysis

Electrophysiology statistics

Data are expressed as mean values $\pm$ standard error of the mean (SEM). For all drug induced changes in DRP, a Student's *t* test was used to compare the group to a chance change in baseline values. If these changes were significantly different than zero, an asterisk is marked directly above or below the bar graph. When 3 or more groups were compared, a one-way ANOVA was performed with Dunnett's or Tukey's multiple comparison post hoc test comparing groups as marked. When two or more groups were compared at multiple stimulus intensities, a two-way ANOVA was completed, and if the interaction between genotype and stimulus intensity was statistically significant, Sidak's multiple comparisons post hoc test was used at each stimulus intensity.

Behavior Statistics

For most behavioral assays, a two- way ANOVA for genotype and sex was used, and a multiple comparison post hoc test was done whenever there was a statistically significant interaction between sex and genotype, as noted in the figure legends. Genital sensitivity

was only run on female mice, and a Mann-Whitney non-parametric students' t test was used to determine significance. For von Frey response rates, a two-way repeated-measures ANOVA was run on combined male and female groups as there was no effect of sex or interaction on their von Frey threshold. Similarly, there was no significant effect/interaction of sex on NORT or rough texture avoidance assays, so male and female mice were combined for statistical analysis. For these behavioral assays, a Student's t test was used to compare the group to no preference. If the group's preference for novel/smooth was significantly different than chance, an asterisk is marked directly above the bar graph. For all behavioral measurements reported in [Figure 6](#), if the two-way ANOVA for either *Grin1*^{+/+} or *Gabrb3*^{+/+} mutants indicated a main effect of sex or interaction between sex and genotype, Sidak's post hoc multiple comparison test was used to determine significant effects of genotype for each sex individually. For aPPI, a two-way ANOVA was done on males and females separately, with the main effect of genotype denoted above the bar graph and Sidak's post hoc multiple comparison test run to determine the statistically significant prepulse intensities. All statistics were done using GraphPad Prism 7.